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OGC® WPS 2.0.2 INTERFACE STANDARD CORRIGENDUM 2

STANDARD
Implementation

APPROVED

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I ABSTRACT

In many cases geospatial or location data, including data from sensors, must be processed before the information can be used effectively. The OGC Web Processing Service (WPS) Interface Standard provides a standard interface that simplifies the task of making simple or complex computational processing services accessible via web services. Such services include well-known processes found in GIS software as well as specialized processes for spatio-temporal modeling and simulation. While the OGC WPS standard was designed with spatial processing in mind, it can also be used to readily insert non-spatial processing tasks into a web services environment.

The WPS standard provides a robust, interoperable, and versatile protocol for process execution on web services. It supports both immediate processing for computational tasks that take little time and asynchronous processing for more complex and time consuming tasks. Moreover, the WPS standard defines a general process model that is designed to provide an interoperable description of processing functions. It is intended to support process cataloguing and discovery in a distributed environment.

II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

TBD, geoprocessing, ogcdoc, OGC document, processes, processing, WPS

III PREFACE

This standard is a continuation of WPS 1.0, a standard for web-based processing of geospatial data. It incorporates a range of change requests that have been submitted since the release of WPS 1.0 and further follows the OGC standard for modular specifications [OGC 08-131r3].

In contrast to the prior version, WPS 2.0 provides a core conceptual model that may be used to specify a WPS in different architectures such as REST or SOAP.

The WPS process model has been encapsulated into separate requirements and conformance classes, so it may be used independently from WPS servers in process catalogs and metadata records. The expressive power of process descriptions has been enhanced by permitting structured (or nested) inputs and outputs. The concept of process profiles has been clarified and extended to support process descriptions at different levels of abstraction.

Conversely, the process model itself has been largely decoupled from the WPS protocol, allowing the use of other domain-specific descriptions of processes, e.g. those defined in SensorML, and to execute them on a WPS server.

This specification also provides a Basic WPS conformance class that comprises the synchronous and asynchronous execution protocol, the WPS process model, and implements HTTP/POST +XML and HTTP/GET+KVP encodings.

Future work will target the definition of process interfaces for common processes based on the process model conformance class. Such profiles will encourage the development of well-defined, reliable, interoperable and exchangeable process implementations.

If OGC baseline and related specifications should further progress towards REST-oriented interfaces, the development of a REST-oriented WPS interface standard should be considered.

IV SECURITY CONSIDERATIONS

No security considerations have been made for this document.

V SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- TU Dresden
- 52°North
- Intergraph Corporation
- Image Matters LLC
- CRP Henri Tudor
- Airbus Defence and Space

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1

SCOPE

1 SCOPE

This document specifies the interface to a general purpose Web Processing Service (WPS). A WPS is a web service that enables the execution of computing processes and the retrieval of metadata describing their purpose and functionality. Typically, these processes combine raster, vector, and/or coverage data with well-defined algorithms to produce new raster, vector, and/or coverage information.

The WPS protocol supports both synchronous and asynchronous execution of processes. Synchronous execution may be used in simple and quick computation scenarios, where the data processing takes little to almost no time. Asynchronous processing is particularly well suited for complex computation scenarios which may take significant time.

The specification uses a core and extensions model to organize its features:

1. A core conceptual model, defining basic requirements for a web based processing service,
2. A process model to support the description and discovery of processes on the web,
3. A basic data model that supports arbitrary (standard or non-standard) data formats for inputs and outputs,
4. A WPS service model and encoding based on OGC baseline standards, and
5. A Dismiss extension to allow clients to terminate asynchronous processing jobs.

2

CONFORMANCE

2 CONFORMANCE

Conformance with this standard shall be checked using all the relevant tests specified in Annex A (normative) of this document. The framework, concepts, and methodology for testing, and the criteria to be achieved to claim conformance are specified in the OGC Compliance Testing Policies and Procedures and the OGC Compliance Testing web site[1].

Table 1 — Conformance classes related to WPS 2.0

CONFORMANCE CLASS	DESCRIPTION	CLAUSE
http://www.opengis.net/spec/WPS/2.0/conf/service/profile/**basic-wps**	Basic WPS service profile	Annex A.1
http://www.opengis.net/spec/WPS/2.0/conf/service/**synchronous-wps**	Synchronous WPS	Annex A.2
http://www.opengis.net/spec/WPS/2.0/conf/service/**asynchronous-wps**	Asynchronous WPS	Annex A.3
http://www.opengis.net/spec/WPS/2.0/conf/**process-model-encoding**	WPS process model encoding	Annex A.4
http://www.opengis.net/spec/WPS/2.0/conf/service/**dismiss-extension**	Dismiss extension	Annex A.6

[1] www.opengeospatial.org/cite

3

NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Arliss Whiteside, Jim Greenwood: OGC 06-121r9, *OGC Web Service Common Implementation Specification*. Open Geospatial Consortium (2010). https://portal.ogc.org/files/?artifact_id=38867.

Policy SWG: OGC 08-131r3, *The Specification Model — Standard for Modular specifications*. Open Geospatial Consortium (2009). https://portal.ogc.org/files/?artifact_id=34762&version=2.

A. Phillips, M. Davis: IETF RFC 4646, *Tags for Identifying Languages*. RFC Publisher (2006). https://www.rfc-editor.org/info/rfc4646.

T. Berners-Lee, R. Fielding, L. Masinter: IETF RFC 3986, *Uniform Resource Identifier (URI): Generic Syntax*. RFC Publisher (2005). https://www.rfc-editor.org/info/rfc3986.

ISO: ISO 8601:2004, *Data elements and interchange formats — Information interchange — Representation of dates and times*. International Organization for Standardization, Geneva (2004). https://www.iso.org/standard/40874.html.

XML Schema Part 2: Datatypes Second Edition, W3C Recommendation 28 October 2004.

4

TERMS AND DEFINITIONS

4 TERMS AND DEFINITIONS

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word "shall" (not "must") is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the 'ModSpec'. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

This document uses the terms defined in Sub-clause 5.3 of [OGC 06-121r9], which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word "shall" (not "must") is the verb form used to indicate a requirement to be strictly followed to conform to this standard.

For the purposes of this document, the following additional terms and definitions apply.

4.1. Process:

A **process** p is a function that for each input returns a corresponding output

$\mu : X \rightarrow Y$

where X denotes the domain of arguments x and Y denotes the co-domain of values y. Within this specification, process arguments are referred to as process inputs and result values are referred to as process outputs. Processes that have no process inputs represent value generators that deliver constant or random process outputs.

4.2. Process description:

A process description is an information model that specifies the interface of a process. A process description is used for a machine-readable description of the process itself but also provides some basic information about the process inputs and outputs.

4.3. Process input:

Process inputs are the arguments of a process and refer to data provided to a process. Each process input is an identifiable item.

4.4. Process output:

Process outputs are the results of a process and refer to data returned by a process. Each process output is an identifiable item.

4.5. Process profile:

A process profile is a description of a process on an interface level. Process profiles may have different levels of abstraction and cover several aspects. On a generic level, a process profile may only refer to the provided functionality of a process, i.e. by giving a verbal or formal definition how the outputs are derived from the inputs. On a concrete level a process profile may completely define inputs and outputs including data type definitions and formats.

4.6. WPS Server:

A WPS Server is a web server that provides access to simple or complex computational processing services.

4.7. Process offering:

A process offering is an identifiable process that may be executed on a particular service instance. A process offering contains a process description as well as service-specific information about the supported execution protocols (e.g. synchronous and asynchronous execution).

4.8. Process execution:

The execution of a process is an action that calculates the outputs of a given process for a given set of data inputs.

4.9. Job:

The (processing) job is a server-side object created by a processing service for a particular process execution. A job may be latent in the case of synchronous execution or explicit in the case of asynchronous execution. Since the client has only oblique access to a processing job, a Job ID is used to monitor and control a job.

4.10. Service profiles for WPS:

A service profile for WPS is a conformance class that defines the general capabilities of a WPS server, by (1) specifying the supported service operations, (2) the process model, (3) the supported process execution modes, (4) the supported operation binding(s).

6

CONVENTIONS

6 CONVENTIONS

This section provides details and examples for any conventions used in the document. Examples of conventions are symbols, abbreviations, use of XML schema, or special notes regarding how to read the document.

6.1. Abbreviated terms

GML	Geography Markup Language
GRS	Coordinate Reference System
HTTP	Hypertext Transfer Protocol
ISO	International Organization for Standardization
KVP	Keyword Value Pair
MIME	Multipurpose Internet Mail Extensions
OGC	Open Geospatial Consortium
UML	Unified Modeling Language
URI	Universal Resource Identifier
URL	Uniform Resource Locator
WPS	Web Processing Service
XML	Extensible Markup Language

6.2. Use of the Term “Process”

The term process is one of the most used terms both in the information and geosciences domain. If not stated otherwise, this specification uses the term process as an umbrella term for any algorithm, calculation or model that either generates new data or transforms some input data into output data as defined in Clause 4.1.

6.3. UML Notation

Unified Modeling Language (UML) static structure diagrams appearing in this specification are used as described in section 5.2 of OGC06-121r9. Further, the following conventions hold:

- UML elements having a package name of “OWS Common” are those defined in the UML model of OWS Common [OGC 06-121r9].
- UML data type Any is used here as an equivalence to XML’s xsd:any.
- UML elements not qualified with a package name are those defined in this standard.

The UML model data dictionary is specified herein in a series of tables. The contents of the columns in these tables are described in section 5.5 of [OGC 06-121r9]. The contents of these data dictionary tables are normative, including any table footnotes.

6.4. Namespace Conventions

The following namespaces are used in this document. The prefix abbreviations used constitute conventions used here, but are not normative. The namespaces to which the prefixes refer are normative, however.

PREFIX	NAMESPACE URI	DESCRIPTION
ows	http://www.opengis.net/ows/2.0	OWS Common 2.0 XML Schema
xlink	http://www.w3.org/1999/xlink	Definitions for XLINK
xml	http://www.w3.org/XML/1998/namespace	XML (required for xml:lang)
xs	http://www.w3.org/2001/XMLSchema	XML Schema

7

WPS CONCEPTUAL MODEL

7 WPS CONCEPTUAL MODEL

The WPS service model defines basic properties of any WPS server. A WPS server is a web service that provides access to pre-defined processes and provides job control operations to instantiate, control and monitor processing jobs (Figure 1).

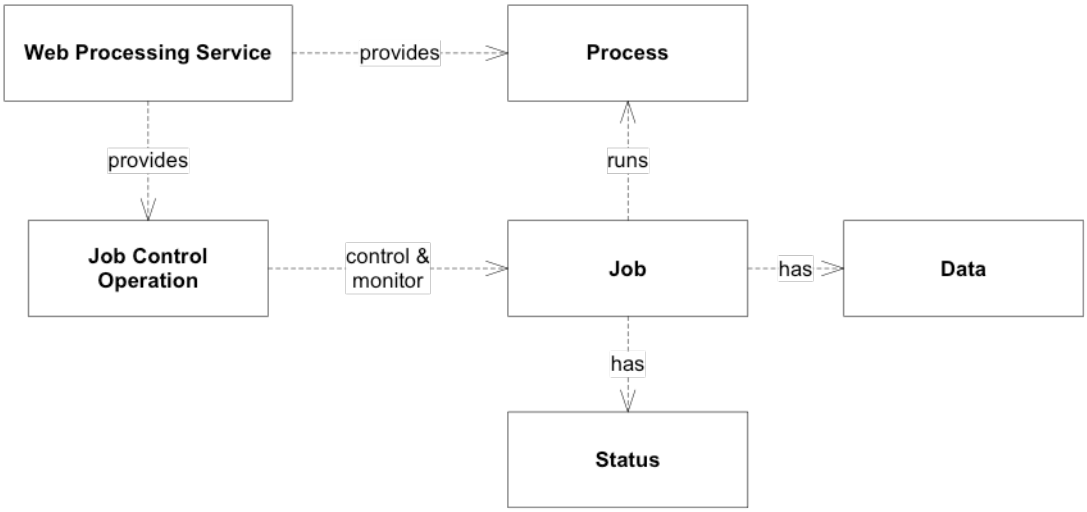

Figure 1 — Artifacts of the WPS service model

REQUIREMENTS CLASS 1	
OBLIGATION	requirement
TARGET TYPE	Derived information model, encoding, and software implementation
NORMATIVE STATEMENTS	Requirement 1-1 Requirements class for service discovery. Requirement 1-2 Requirements class for service capabilities. Recommendation 1-1 Requirements class for supported process models. Requirement 1-3 Requirements class for job control. Requirement 1-4 Requirements class for process execution. Requirement 1-5 Requirements class for data transmission between service and client. Requirement 1-6 Requirements class for job monitoring.

7.1. Service Discovery

Any WPS server shall be self-contained, i.e. provide an initial endpoint that can be used by a WPS client to determine the server's capabilities.

REQUIREMENTS CLASS 2	
OBLIGATION	requirement
TARGET TYPE	Derived information model, encoding, and software implementation
NORMATIVE STATEMENTS	Requirement 2-1 All WPS servers shall have an initial end-point (HTTP URI). Requirement 2-2 The service shall provide a systematic discovery mechanism for all service capabilities. Requirement 2-3 The discovery mechanism for the service capabilities shall be predictable from the initial endpoint.

7.2. Service Capabilities

The basic capabilities of any WPS server fall into two categories: The first category comprises capabilities for process discovery and retrieval of process descriptions. The second category comprises capabilities to manage and monitor processing jobs.

Since the processes provided by a WPS server may have different degrees of complexity, the server shall indicate the allowed job control capabilities mode per process offering.

Further service capabilities, i.e. for secure communication and user authentication may be provided with the service but are neither covered nor restricted by this specification as long as they do not alter or change the semantics of other job control capabilities.

REQUIREMENTS CLASS 3	
OBLIGATION	requirement
TARGET TYPE	Derived information model, encoding, and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process
NORMATIVE STATEMENTS	Requirement 3-1 The service shall provide a process offering capabilities. This capability informs service clients about the available processes.

REQUIREMENTS CLASS 3
Requirement 3-2 All process offerings shall provide an identifier for their process model used. Requirement 3-3 The service shall provide job control and monitoring capabilities. These capabilities enable service clients to manage processing jobs via the service interface. Requirement 3-4 The service shall indicate the allowed job control capabilities per process.

7.3. Abstract Process Model

The abstract process model specifies generic requirements for process offerings that can be used in conjunction with WPS. Processes, as well as their inputs and outputs, are elements with identity. A process input may have arbitrarily defined value cardinality, i.e. for a given input, multiple datasets may be submitted for execution. A process output always has a value cardinality of one. Process inputs and outputs may also be nested. Identifiers for inputs and outputs shall be unique. For nested children it is sufficient to have a unique nesting path. Any input and output that does not have child elements shall have a defined data type so that a client is aware of the valid data formats for process execution.

The abstract process model provides many degrees of freedom for process descriptions. However, it does not enforce additional complexity for very simple processes.

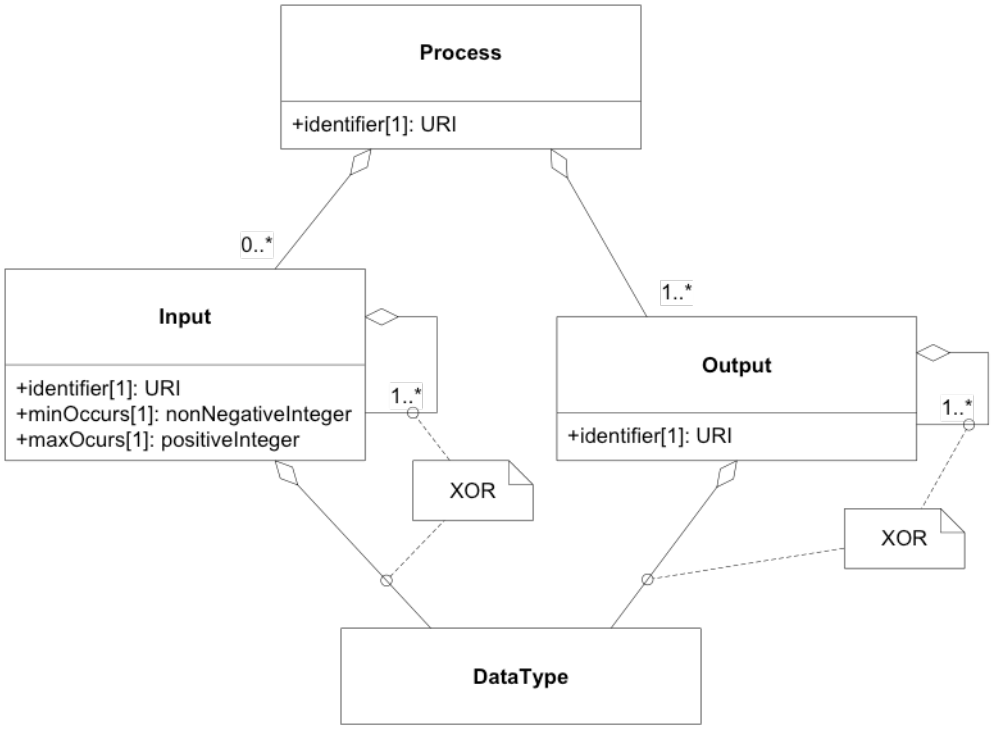

Figure 2 — Abstract process model UML class diagram

REQUIREMENTS CLASS 4	
OBLIGATION	requirement
TARGET TYPE	Derived information model, encoding, and software implementation
NORMATIVE STATEMENTS	Requirement 4-1 *A process shall provide a unique identifier to distinguish the process from other processes.* Requirement 4-2 *The unique identifier of a process shall be a character string or preferably a resolvable URI.* Requirement 4-3 *A process may have an arbitrary number of inputs (zero or more).* Requirement 4-4 *Each input of a process shall have a unique identifier to distinguish the input from all other inputs.* Requirement 4-5 *The unique identifier of an input shall be a character string.* Requirement 4-6 *Each process input shall have a defined cardinality for values.* Requirement 4-7 *Process outputs may be nested according to Figure 2.* Requirement 4-8 *The identifier of a nested input shall be unique within its nesting node.* Requirement 4-9 *A process shall have one or more outputs.* Requirement 4-10 *Each output of a process shall have a unique identifier to distinguish the output from all other outputs of the process.* Requirement 4-11 *The unique identifier of an output shall be a character string.* Requirement 4-12 *Each process output shall have a value cardinality of one.* Requirement 4-13 *Process outputs may be nested according to Figure 2.* Requirement 4-14 *The identifier of a nested output shall be unique within its nesting node.* Requirement 4-15 *All inputs and outputs that do not serve as nesting parents shall have a defined data format.* Requirement 4-16 *If inputs and outputs require encoding attributes, these shall be limited to the properties defined in Table 2.*

Table 2 — Data encoding properties

STATUS	DEFINITION
mimetype	Media type of the data.
encoding	Encoding procedure or character set used (e.g. raw, base64, or UTF-8).
schema	Identification of the data schema.

7.4. Job Control

The execution capability, that permits WPS clients to instantiate and run processing jobs, is the most prominent job control capability. Furthermore, the ability to dismiss or delete a job is useful for long-running processes in order to release server resources.

REQUIREMENTS CLASS 5	
OBLIGATION	requirement
TARGET TYPE	Derived information model, encoding, and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/capabilities
NORMATIVE STATEMENTS	Requirement 5-1 *The service shall assign a unique identifier to each job.* Requirement 5-2 *The Service shall return an exception, if the client has attempted to use an invalid job identifier.* Requirement 5-3 *The service shall provide the ability to execute a process, i.e. to create a new job for a given process execution capability. This capability enables service clients to execute the process defined in capabilities.* Requirement 5-4 *The service may provide the capability to dismiss a submitted job. This capability enables service clients to indicate that they are no longer interested in the job or its results and permit the server to free the associated resources as soon as possible.*

7.5. Process Execution

Process executions on a WPS server may be run either synchronously (Figure 3) or asynchronously (see Figure 4). Synchronous execution is a suitable approach for jobs that take a relatively short time[2] to complete. Asynchronous execution is preferable for jobs that may take a long time to complete.

In the synchronous case, a WPS client submits an execute request to the WPS server and keeps listening for a response until the processing job has completed and the processing result has been returned. This requires a persistent connection between client and server.

[2]Short and long are relative terms and requires considerations on both client and server side and the application context. This specification does not give any guidelines regarding what short or long means. There may also be an architectural consideration that suggests either synchronous or asynchronous communication.

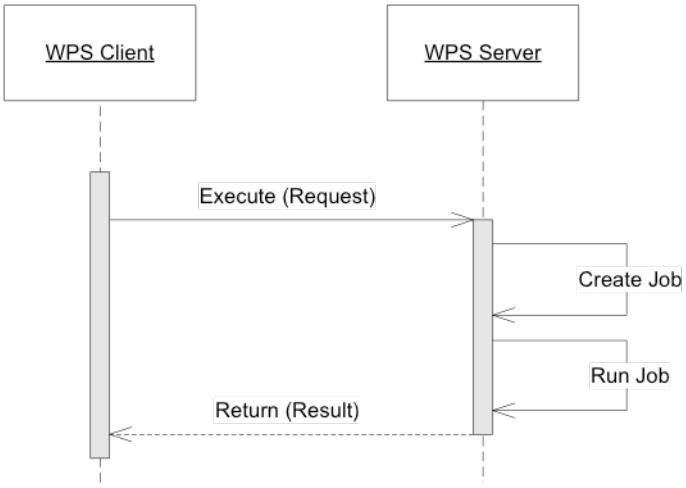

Figure 3 — Synchronous process execution UML sequence diagram

In the asynchronous case, the client sends an execute request to the WPS server and immediately receives a status information response. This information confirms that the request was received and accepted by the server and that a processing job has been created and will be run in the future. The status information response also contains a processing job identifier that is used by the client when checking to see if execution has completed. (The client may also manage the processing job using available steering capabilities.) In addition, the status information response contains the result location, i.e. the URL where the processing result can be found after the processing job has completed.

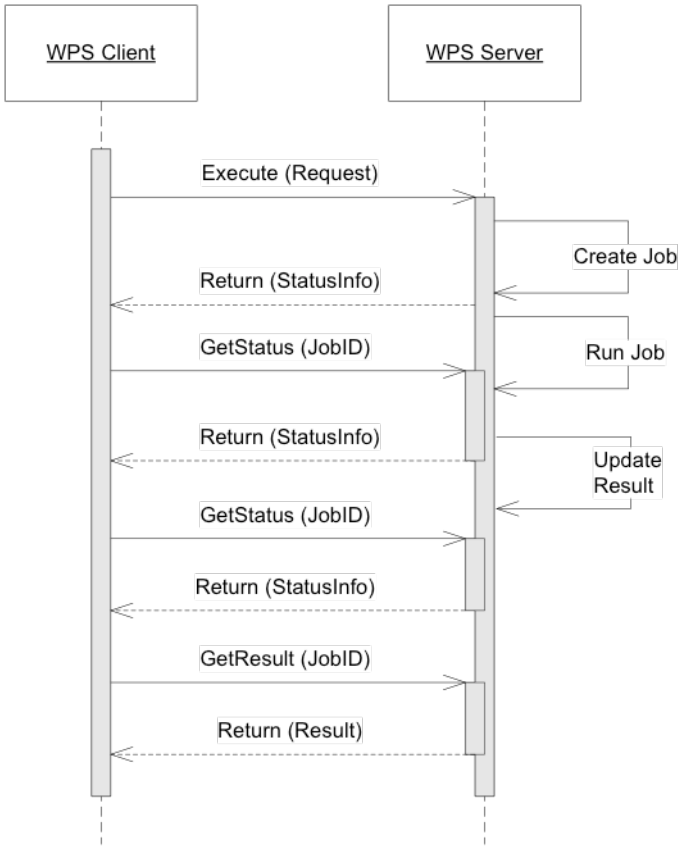

Figure 4 — Asynchronous process execution UML sequence diagram

REQUIREMENTS CLASS 6	
OBLIGATION	requirement
TARGET TYPE	Derived information model, encoding, and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/job-control
NORMATIVE STATEMENTS	Requirement 6-1 *The service shall permit a client to determine the process to be executed.* Requirement 6-2 *The service shall return an exception if the client has attempted to execute an unavailable process.* Requirement 6-3 *On successful execution, the service shall deliver a result to the client. This result will contain the output data or a reference to the data.* Requirement 6-4 *The execute result may have an expiration date. After that time the output data will be inaccessible for the client.* Requirement 6-5 *On failed execution, the service shall deliver an exception to the client.* Requirement 6-6 *The service shall permit a client to specify the input data to be used for a process execution.* Requirement 6-7 *The service shall return an exception if the client has specified invalid input data for process execution.*

REQUIREMENTS CLASS 6
Requirement 6-8 *The service shall permit a client to define the desired data exchange format for the results of the process execution, based on the advertised formats for the process offering.* Requirement 6-9 *The service shall return an exception if the client has defined an unsupported data exchange format for the results.* Requirement 6-10 *The service shall provide execution capabilities in synchronous, asynchronous or both modes.* Requirement 6-11 *The service shall indicate the allowed execution modes per process offering.* Requirement 6-12 *In the case of multiple supported execution modes, the WPS server shall permit the client to specify the desired execution mode.* Requirement 6-13 *If the client does not specify the desired execution mode, the service shall automatically pick[3] an appropriate[4] mode out of the supported modes.* Requirement 6-14 *The service shall return an exception if the client has specified an unsupported execution mode.*

7.6. Data Transmission by Value and by Reference

Data exchange between WPS clients and servers requires an agreement on the general data exchange patterns and suitable communication protocols. This specification defines two general data transmission modes for data exchange between WPS client and server.

Clients may send input data to or receive output data from a process in two distinct ways: (1) by reference, and (2) by value (see Figure 5). For brevity, this figure only shows the transmission patterns in the pure form, i.e. the same pattern is used for all inputs and outputs. However, mixed patterns are possible. Typically, small or atomic data such as integers, doubles or short strings are submitted by value. Large data inputs (outputs) are usually supplied by reference.

[3]The duration of the execution may also depend on data size, network bandwidth, and available computing resources on the server side. In general, the client is not aware of these parameters, and may choose to defer the decision to the server.

[4]Synchronous execution is generally preferred for very quick computations while asynchronous execution shall be chosen for long-running jobs. Foremost, the service shall respond quickly to the client to avoid connection timeouts.

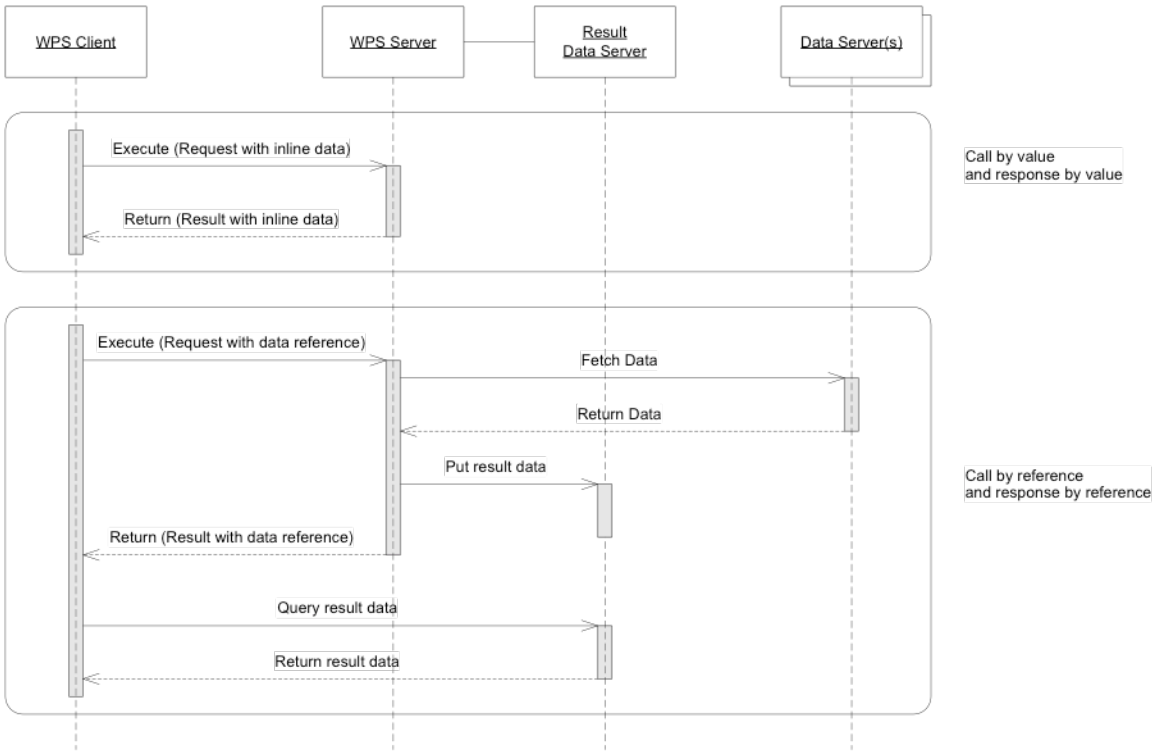

Figure 5 — Execute call and response "by value" and "by reference" UML sequence diagram.

This specification makes no assumptions about the encoding of the transmitted data. Due to the variety of data formats and supplementary encoding procedures, the receiver may not be able to automatically detect the data format. The set of encoding attributes that may be used by the receiver to decode inline or directly referenced data is defined by the data format description (see Table 2).

REQUIREMENTS CLASS 7	
OBLIGATION	requirement
TARGET TYPE	Derived information model, encoding, and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process-execution
NORMATIVE STATEMENTS	Requirement 7-1 *The service shall accept input data by value.* Requirement 7-2 *The service shall accept input data by reference.* Requirement 7-3 *The service shall return an exception, if the given input data reference is inaccessible.* Requirement 7-4 *The service shall support output data delivery by value, and / or by reference.* Requirement 7-5 *The supported output modes shall be specified per process offering.* Requirement 7-6 *In the case of multiple available data transmission modes (by value AND by reference), the Service shall allow a client to specify the desired data-transmission mode for each output.*

REQUIREMENTS CLASS 7	
	Requirement 7-7 *The service shall return an exception if the client has specified an unsupported transmission mode.*

7.7. Job Monitoring

By definition, a processing job is a server-side object created by a processing service in response for a particular process execution. It is comprised of a process definition (i.e. one of the process offerings defined in the WPS server's capabilities), the input data provided or specified by the WPS client, and the outputs that are eventually delivered when the job has been completed.

Since the processing job is a server-side object, a WPS client has no means to inspect the status of a job on its own. Therefore, the server should provide a unique identifier for each job. For privacy, it is recommended to keep this identifier confidential between client and server.

For jobs running in asynchronous mode, the WPS server shall also provide monitoring information and it may also contain estimates on the completion time or additional elements related to status polling.

If a client finds a polling time in the status information, it shall respect it and behave accordingly. The service may rely on the client polling roughly around this time to obtain updated status information.

If a client finds an expiry date in the status information, it shall respect it and behave accordingly, i.e. ensure that the execution result is evaluated on time and the outputs are retrieved before the job is removed from the server. This requirement allows for robust WPS implementations and the timely re-allocation of server resources.

This section also defines a basic status set to communicate the status of a server-side job to the client. Extensions of this specification may introduce additional states for fine-grained monitoring or domain-specific purposes.

REQUIREMENTS CLASS 8	
OBLIGATION	requirement
TARGET TYPE	Derived information model, encoding, and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/job-control http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process-execution
NORMATIVE STATEMENTS	Requirement 8-1 *The Service shall provide access to machine-readable status information for asynchronously running jobs.* Requirement 8-2 *In general, the service shall use the terms of the basic status set (Table 3) describe the current status of a processing job.*

REQUIREMENTS CLASS 8	
	Requirement 8-3 *The Service may report the progress of a running job in percent.* Requirement 8-4 *The Service may provide an estimate when the process results will be available.* Requirement 8-5 *The Service may suggest a polling time for the next status check.* Requirement 8-6 *The server may report an expiration time for a job. After this time, a job identifier will become invalid and existing results are removed from the server.*

Table 3 — Basic status set for jobs

STATUS	DEFINITION
Succeeded	The job has finished with no errors.
Failed	The job has finished with errors.
Accepted	The job is queued for execution.[a]
Running	The job is running.[a]

[a] States are only defined for asynchronous execution.

8

WPS NATIVE PROCESS MODEL

8 WPS NATIVE PROCESS MODEL

The WPS native process model is a realization of the abstract process model defined in Clause 7.3. The native process model breaks down into the following components:

1. A general structure for process interface descriptions (Clause 8.1, Clause 8.4) that support process discovery and cataloging
2. A set of data types to submit and receive data during process execution (Clause 8.2, Clause 8.3)
3. A framework for defining process profiles to improve interoperability between different process implementations (Clause 8.5).

The native process model is encapsulated in a separate conformance class (Annex A.4) and can be used independently from the rest of the WPS specification. It is designed to provide an interoperable description of processing functions, regardless whether they are offered by a web service for remote invocation, or reside in a Desktop GIS for local use. By abstracting from a concrete implementation this process model may be used to create metadata records in process catalogs or help comparing and aligning the processing functions in different systems.

This section describes the information model of requirements. The corresponding XML and plain text encodings are specified in Clause 9.

REQUIREMENTS CLASS 9	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process
NORMATIVE STATEMENTS	Requirement 9-1 *Requirements class for the common description type.* Requirement 9-2 *Requirements class for IO format.* Requirement 9-3 *Requirements class for data types.* Requirement 9-4 *Requirements class for process description.* Requirement 9-5 *Requirements class for process profile.*

8.1. Common Description Type

Descriptive elements of processes, inputs and outputs are derived from the BasicIdentificationType provided by OWS Common (Figure 6). Other descriptive information

shall be recorded in the Metadata element in the form of simple links with an appropriate role identifier.

REQUIREMENTS CLASS 10	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process OWS Common 2.0 — BasicDescriptionType
NORMATIVE STATEMENTS	Requirement 10-1 *Process descriptions as well as the associated process inputs and outputs shall be derived from the OWS Common BasicIdentificationType.* Requirement 10-2 *The DescriptionType shall comply with the structure defined in Figure 6 and Table 4.* Requirement 10-3 *For linking documentation material or other metadata with the process itself, its inputs and its outputs, the metadata element from the BasicIdentificationType shall be used.* Requirement 10-4 *The metadata elements within process descriptions shall be simple links with a human-readable title (Table 5).* Requirement 10-5 *The metadata elements for documentation shall use the role identifier "http://www.opengis.net/spec/wps/2.0/def/process/description/documentation".*

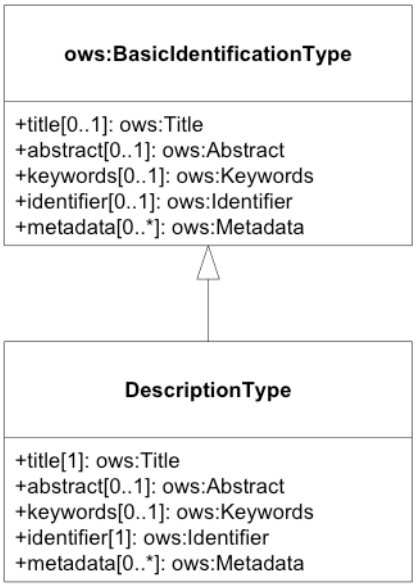

Figure 6 — DescriptionType for processes, process inputs and process outputs UML class diagram

Table 4 — Properties of the DescriptionType structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Title	Title of the process, input, and output. Normally available for display to a human.	ows:Title	One (mandatory)
Abstract	Brief narrative description of a process, input, and output. Normally available for display to a human.	ows:Abstract	Zero or one (optional) Include when available and useful.
Keywords	Keywords that characterize a process, its inputs, and outputs.	ows:Keywords	Zero or more (optional) Include when available and useful.
Identifier	Unambiguous identifier of a process, input, and output.	ows:Identifier Value is a URI or HTTP-URI[a]	One (mandatory)
Metadata	Reference to additional metadata about this item.	ows:Metadata Allowed values are specified in Table 5.	Zero or more (optional)

[a] Additional content such as separate code space and version attributes in the Identifier element are not allowed.

Table 5 — Properties of the Metadata structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Title	Title of the documentation. Normally available for display to a human.	Character String	One (mandatory)
Link type	Type of the xlink, fixed to simple.	Character String, fixed to “simple”.	One (mandatory)
Role	Role identifier, indicating the role of the linked document.	HTTP-URI	One (mandatory)
href	Reference to a documentation site for a process, input, or output.	HTTP-URI	One (mandatory)

8.2. Data Description Structure

The DataDescription structure contains basic properties for defining data inputs and outputs, including mimetype, encoding and schema. These properties specify supported formats for input and output data of computing processes. Any input or output item may support multiple

formats, one of which is the default format. Processes may require that an input or output data set does not exceed a certain data volume.

REQUIREMENTS CLASS 11	
OBLIGATION	requirement
TARGET TYPE	Derived information model, encoding, and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process
NORMATIVE STATEMENTS	Requirement 11-1 *Format descriptions of process inputs and outputs shall comply with the DataDescription structure defined in Figure 7 and Table 6.* Requirement 11-2 *One of the formats defined in the DataDescription structure shall be the default format, i.e. have the attribute "default" set to "true".*

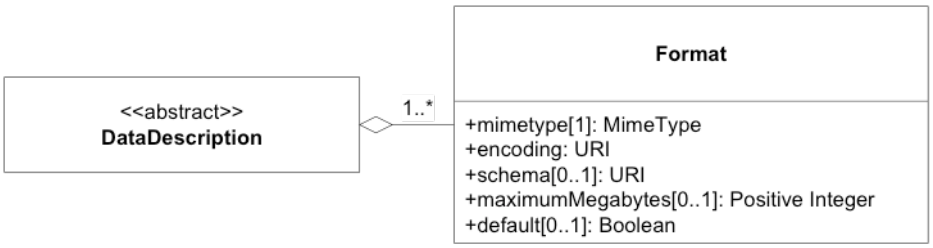

Figure 7 — DataDescription and supported formats UML class diagram

Table 6 — Format properties

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
mimetype	Media type of the data.	Character String	One (mandatory)
encoding	Encoding procedure or character set of the data (e.g. raw or base64)	Character String, fixed to "simple".	One (mandatory)
schema	Identification of the data schema.	HTTP-URI	One (mandatory)
maximum Megabytes	The maximum size of the input data, in megabytes.	Integer	Zero or one (optional)
default	Indicates that this format is the default format.[a]	Boolean	Zero or one (conditional)[a],[b]

[a] Defaults to FALSE if omitted.

[b] One of the formats included in the DataDescription structure shall have the attribute "default" set to "true".

8.3. Data Types

This specification defines three common data types for process input and output data (Figure 8):

1. ComplexData, such as GML or geo-referenced imagery. This type is kept generic regarding the content and may be extended to provide more detailed domain-specific information.
2. LiteralData, defined as a value with an optional unit.
3. BoundingBoxData, defined as a minimum bounding rectangle in geographic coordinates.

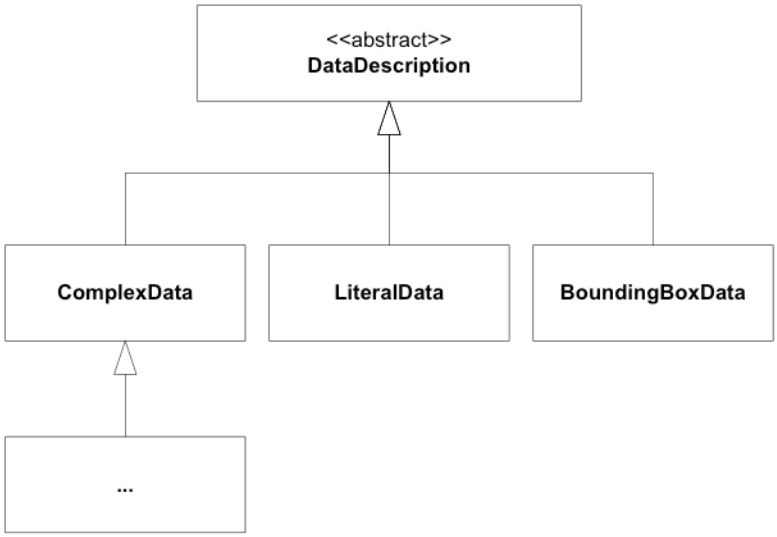

Figure 8 — I/O data types overview

REQUIREMENTS CLASS 12	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process http://www.opengis.net/spec/WPS/2.0/req/native-process/model/io-format OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 12-1 *Requirements class for the complex data.* Requirement 12-2 *Requirements class for the literal data.* Requirement 12-3 *Requirements class for the bounding box data.*

8.3.1. Complex data

The ComplexData type does not describe the particular structure for value encoding. Instead, the passed values must comply with the given format and the extended information, if provided.

REQUIREMENTS CLASS 13	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 13-1 *equirements class for the complex data description.* Requirement 13-2 *Requirements class for the complex data values.*

8.3.1.1. ComplexData Description

The ComplexData structure is a direct realization of the abstract DataDescription element (Figure 7). It relies on the encoding attributes defined in Table 6 for a basic description of input and output data. In cases where these attributes do not sufficiently capture the structure and content of the required or produced data, the ComplexData element provides the ability to include any other descriptive elements next to the format specification. This hook may be used by process providers and consumers to communicate essential information and further constraints for input and output data items.

REQUIREMENTS CLASS 14	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model http://www.opengis.net/spec/WPS/2.0/req/native-process/model/io-format OWS Common 2.0
NORMATIVE STATEMENT	Requirement 14-1 *The description of valid complex data input and output shall comply with the structure defined in Table 7.*

Table 7 — ComplexData description properties

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Format	Identifies a valid format for an input or output.	Format properties, see Table 6.	One or more (mandatory)
Any	Placeholder for schema extensions to WPS complex data.	Any type.	Zero or more (optional)

8.3.1.2. ComplexData Values

A Complex data value is directly passed to (or returned by) a process. The generic nature of Complex data does not permit a particular structure for value encoding. Instead, this structure is defined by the ComplexData description and the passed values must comply with the given format and the extended information, if provided.

REQUIREMENTS CLASS 15	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model http://www.opengis.net/spec/WPS/2.0/req/native-process/model/datatypes/complex-data/description
NORMATIVE STATEMENT	Requirement 15-1 *Complex data values must comply with the encoding specified in the complex data description.*

8.3.2. Literal Data

The LiteralData type encodes atomic data such as scalars, linear units, or well-known names. Domains for LiteralData are a combination of data types (e.g. Double, Integer, String), a given value range, and an associated unit (e.g. meters, degrees Celsius).

REQUIREMENTS CLASS 16	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process OWS Common 2.0

REQUIREMENTS CLASS 16	
NORMATIVE STATEMENTS	Requirement 16-1 *Requirements class for the literal data description.* Requirement 16-2 *Requirements class for the literal data values.*

8.3.2.1. LiteralData Description

The LiteralData description structure inherits essential elements from ows:DomainType allowing it to specify value domains including Units of Measure and Default Values. It restricts ows:DomainType by forbidding:

1. "NoValues" for a particular domain
2. the ability to specify further metadata on the values (since this information is already present at the level of input and output definitions in the DescriptionType element).

LiteralData data types should use the well-known types from XML Schema by their URI definition[5]. Table 11 lists the recommended URIs for the most common literal data types.

REQUIREMENTS CLASS 17	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model http://www.opengis.net/spec/WPS/2.0/req/native-process/model/io-format OWS Common 2.0
NORMATIVE STATEMENT	Requirement 17-1 *The description of valid literal data input and output shall comply with the structure defined in Figure 9, Table 8, Table 9, Table 10, and Table 11.*

[5]see http://www.w3.org/TR/xmlschema-2/#built-in-datatypes

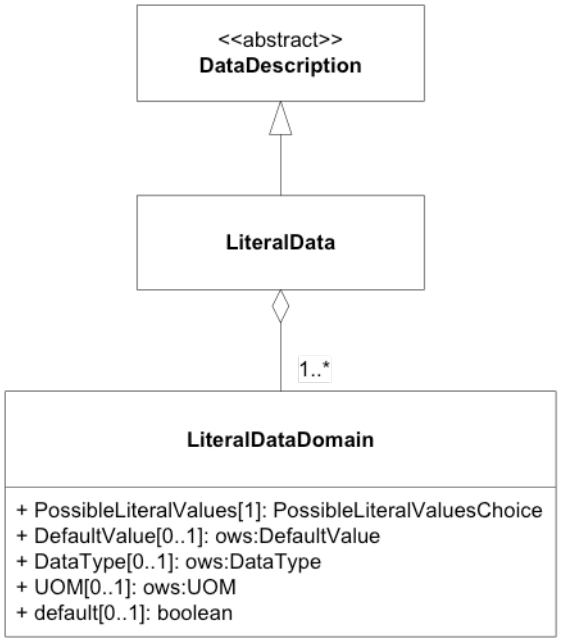

Figure 9 — LiteralData UML class diagram

Table 8 — The LiteralData structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Format	Identifies a valid format for an input or output.	Format properties, see Table 6.	One or more (mandatory)
LiteralDataDomain	The valid domain for literal data	LiteralDataDomain type	One or more (mandatory)

Table 9 — Parts of the LiteralDataDomain structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
PossibleLiteral Values	Identifies a valid format for an input or output.	PossibleLiteralValuesChoice, see Table 10.	One (mandatory)
DataType	Reference to the data type of this set of values	ows:DataType. The use of well-known data type URNs is highly recommended; see Table 11.	One (mandatory)
UOM	Indicates that this quantity has units and provides the unit of measurement.	ows:ValuesUnit	Zero or one (optional) Include when values have units or reference system.

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
DefaultValue	Default value for this quantity.	ows:DefaultValue	Zero or one (optional) Include if there is a default.[a]
default	Indicates that this is the default/native domain.	Boolean, defaults to false.	Zero or one (conditional)[b],[c]

[a] For outputs, the DefaultValue has no meaning and shall thus be omitted.

[b] Defaults to FALSE if omitted.

[c] One of the formats included in the LiteralData structure shall have the attribute "default" set to "true".

Table 10 — Parts of the PossibleLiteralValuesChoice structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
AllowedValues	List of all valid values and/or ranges of values for this quantity.	ows:AllowedValues	Zero or one (conditional)[a]
AnyValue	Specifies that any value is allowed for this quantity.	ows:AnyValue	Zero or one (conditional)[a]
Values Reference	Reference to list of all valid values and/or ranges of values for this quantity.	ows:ValuesReference	Zero or one (conditional)[a]

[a] One and only one of these three items shall be included.

Table 11 — Recommended data type URIs for literal data

DATA TYPE	URI
String	http://www.w3.org/2001/XMLSchema#string
Integer	http://www.w3.org/2001/XMLSchema#integer
Decimal	http://www.w3.org/2001/XMLSchema#decimal
Boolean	http://www.w3.org/2001/XMLSchema#boolean
Double	http://www.w3.org/2001/XMLSchema#double
Float	http://www.w3.org/2001/XMLSchema#float

8.3.2.2. LiteralData Values

LiteralData values represent values that correspond to a particular domain defined in the LiteralData structure. Figure 10 shows the mapping from the LiteralValue structure to the corresponding elements in the LiteralDataDescriptionType.

REQUIREMENTS CLASS 18	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model http://www.opengis.net/spec/WPS/2.0/req/native-process/model/datatypes/literal-data/description OWS Common 2.0
NORMATIVE STATEMENT	Requirement 18-1 *The description of valid literal data values shall comply with the structure defined in Figure 10 and Table 12.*

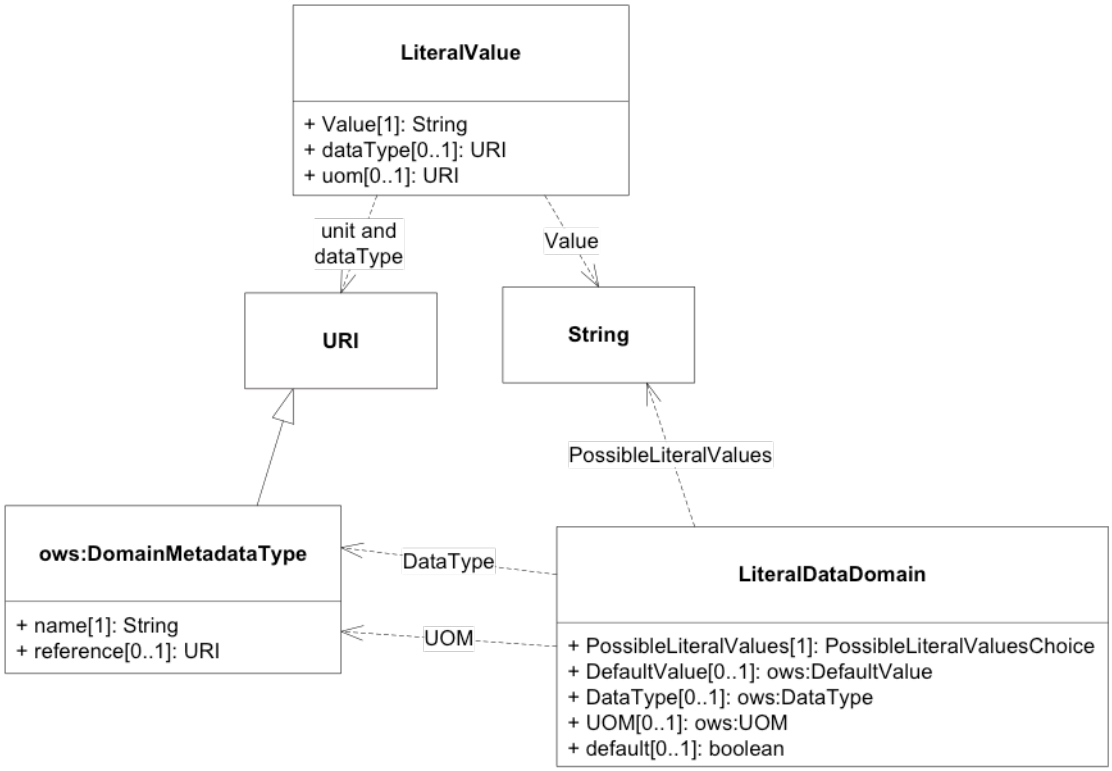

Figure 10 — LiteralValue UML class diagram

Table 12 — Parts of the LiteralValue structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Value	String representation of the actual value.	Character String	One (mandatory)
dataType	The data type of the Value.	URI	Zero or one (optional)[a]
uom	The unit of measurement of the value.	URI	Zero or one (optional)[a]

[a] If not specified, the relevant defaults from the LiteralData description (see Clause 8.3.2.1) will be used.

8.3.3. BoundingBox Data

Bounding box data serves a variety of purposes in spatial data processing. Some simple applications are the definition of extents for a clipping operation or the definition of an analysis region. This specification inherits the bounding box specification from OWS Common.

REQUIREMENTS CLASS 19	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process OWS Common 2.0
NORMATIVE STATEMENT	Requirement 19-1 *Requirements class for the bounding box data description.* *Requirements class for the bounding box data values.*

8.3.3.1. BoundingBox Description

The domain for bounding box data is described by a listing of supported CRSs.

REQUIREMENTS CLASS 20	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model http://www.opengis.net/spec/WPS/2.0/req/native-process/model/io-format OWS Common 2.0

REQUIREMENTS CLASS 20	
NORMATIVE STATEMENT	Requirement 20-1 *The description of valid bounding box data input and output shall comply with the structure defined in Figure 11, Table 13, and Table 14.*

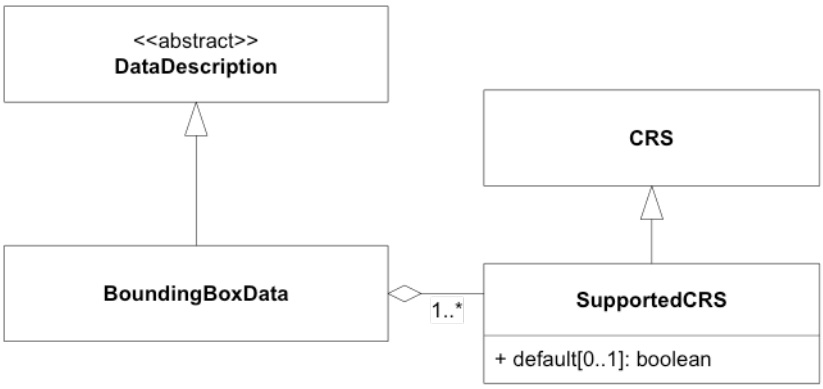

Figure 11 — BoundingBoxData UML class diagram

Table 13 — The BoundingBox structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Format	Identifies a valid format for an input or output.	Format properties, see Table 6.	One or more (mandatory)
Supported CRS	The supported CRS for BoundingBox data.	SupportedCRS type, see Table 14.	One or more (mandatory)

Table 14 — The SupportedCRS type structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
CRS	Reference to a CRS definition.	URI	One or more (mandatory)
default	Indicates that this CRS is the default CRS.	Boolean, defaults to false.	Zero or one (conditional)[a],[b]

[a] Defaults to FALSE if omitted.

[b] One of the formats included in the BoundingBox structure shall have the attribute “default” set to “true”.

8.3.3.2. BoundingBox Values

Values for bounding boxes are specified in the BoundingBox data type from OWS Common [OGC 06-121r9]. For consistency with the BoundingBoxData description, the specification of a CRS is mandatory.

REQUIREMENTS CLASS 21	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model http://www.opengis.net/spec/WPS/2.0/req/native-process/model/datatypes/bounding-box-data/description OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 21-1 *The description of bounding box data values shall comply with the structure defined in OWS Common [OGC 06-121r9].* Requirement 21-2 *The CRS component of the Bounding box values structure shall not be empty.*

8.4. Process Description

This section defines information structures that describe a process. It includes elements that link to documentation resources on the behavior and mechanics of a process as well as descriptive elements about its inputs and outputs. The process description model realizes and extends the requirements defined in the abstract process model in Clause 7.3.

A process description is an extension of the DescriptionType (Figure 12). It shall be used to express identifier, title, and abstract and to link to associated metadata elements that provide additional or more detailed information about the process. An additional language attribute shall be used to indicate the language of human readable elements in the description of the process and its inputs and outputs.

The description structures for process inputs and outputs inherit common elements from the DescriptionType (Clause 8.1). These elements shall be used to express identifier, title, and abstract and to link to associated metadata elements that provide additional or more detailed information about the process inputs and outputs. The content of human readable elements in the description of inputs and outputs shall adhere to the language indicated in the process description.

Process inputs are arguments to a process. Process inputs have a cardinality in order to (1) pass multiple values with the same identifier to a process, or (2) declare process inputs as optional

(cardinality "0"). Input elements may be simple (i.e. the input has no sub-inputs attached) or aggregate (i.e. the input has one or more sub-input elements attached). A simple input includes a realization of the DataDescription element. An aggregate input contains one or more sub-inputs.

Outputs are the return values of a process. Outputs have a cardinality of one. Output elements may be simple (i.e. the output has no sub-outputs attached) or aggregate (i.e. the output has one or more sub-output elements attached). A simple output includes a realization of the DataDescription element. An aggregate output contains one or more sub-outputs.

REQUIREMENTS CLASS 22	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process http://www.opengis.net/spec/WPS/2.0/req/native-process/model/description-type IETF RFC 4646
NORMATIVE STATEMENTS	Requirement 22-1 *The process description shall be in compliance with the WPS process model requirements.* Requirement 22-2 *A process description shall comply with the structure defined in Figure 12 and Table 15.* Requirement 22-3 *The language of the human-readable elements within the process description shall be identified by a language identifier as specified in IETF RFC 4646.* Requirement 22-4 *The description of process inputs and outputs shall comply with the structure defined in Figure 12, Table 16, and Table 17.*

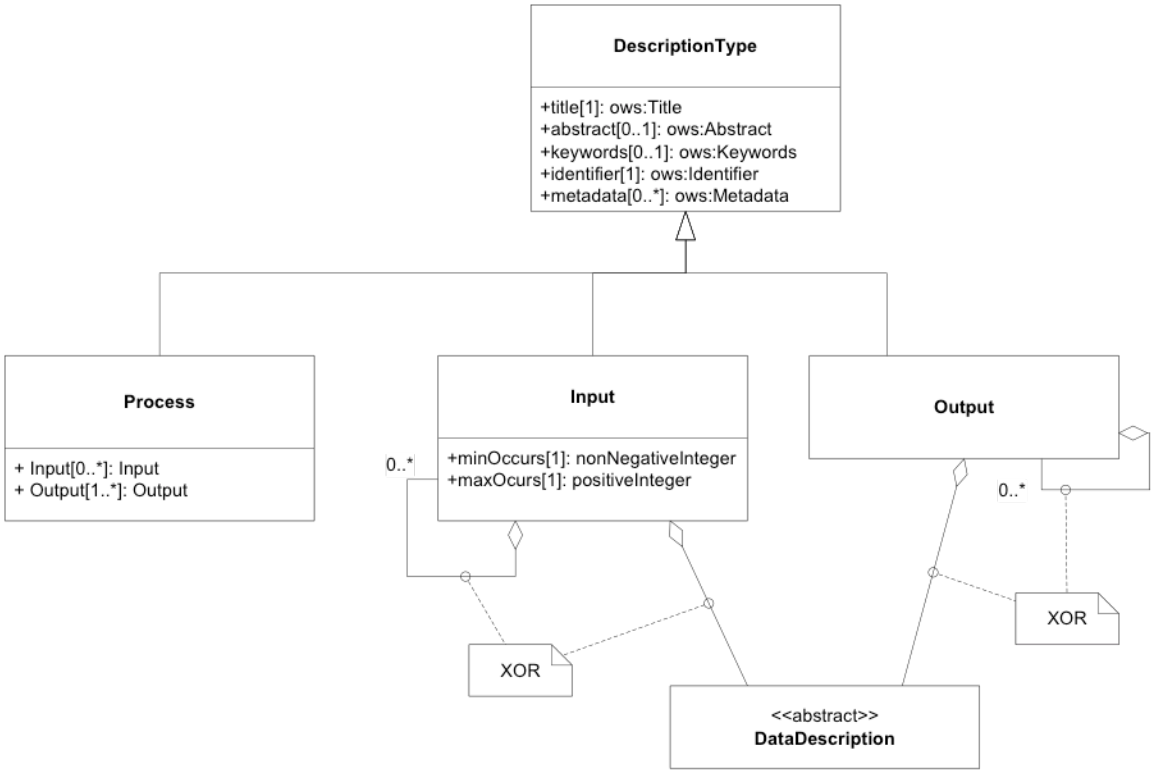

Figure 12 — Process UML class diagram

Table 15 — The Process structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Title	Inherited from Table 4		
Abstract			
Identifier			
Metadata			
Language	Language identifier for the human readable process description elements.	Character String. This language identifier shall be as specified in IETF RFC 4646.	One (mandatory)
Input	Input items (arguments) of a process.	Input structure, see Table 16.	Zero or more (optional)
Output	Output items (results) of a process	Output structure, see Table 17.	One or more (mandatory)

Table 16 — Parts of the Input structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Title			
Abstract			
Keywords	Inherited from Table 4		
Identifier			
Metadata			
minOccurs[a]	Minimum number of times that values for this parameter are required	Non-negative integer; defaults to "1", '0' means the input is optional.	Zero or on (optional)
maxOccurs[a]	Maximum number of times that this parameter may be present	Non-negative integer, defaults to "1".	Zero or one (optional)
Data Description	Data type and domain of this input.	A realization of DataDescription, i.e. ComplexData, LiteralData, Bounding BoxData.	Zero or one (conditional)[b]
Input	Nested Input.[c]	Input structure, Table 16 (this table).	Zero or more (conditional)[b]

[a] The minOccurs and maxOccurs parameters have identical semantics to the like-named XML Schema occurrence constraints.

[b] The input shall either include one realization of DataDescription or an arbitrary number of sub-Inputs.

[c] It is recommended to keep the nesting level as low as possible.

Table 17 — Parts of the Output structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Title			
Abstract	Inherited from Table 4		
Keywords			
Identifier			

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Metadata			
Data Description	Data type and domain of this input.	A realization of DataDescription, i.e. ComplexData, LiteralData, BoundingBoxData.	Zero or one (conditional)[a]
Output	Nested Output.[b]	Output structure, Table 17 (this table).	Zero or more (conditional)[a]

[a] The output shall either include either one realization of DataDescription or an arbitrary number of sub-Outputs.

[b] It is recommended to keep the nesting level as low as possible.

8.5. Process Profiles

Process profiles are blueprints for process implementations and are meant to harmonize process implementations to a certain degree. They serve as a reference for process implementations by providing a description of what the process actually does. While this specification does not attempt to enforce or suggest any particular process profiles, it provides a mechanism to define common processing functionality within the scope of WPS, thus supporting basic process cataloguing and retrieval tasks for distributed processing infrastructures. Depending on the degree of harmonization, the definitions of process profiles may be used to foster a common understanding of widely used processing functions. However, they may also be used to harmonize the technical details of process interfaces and thus document particular interoperability arrangements between process providers and consumers.

REQUIREMENTS CLASS 23	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process
NORMATIVE STATEMENTS	Requirement 23-1 *Requirements class for process concepts.* Requirement 23-2 *Requirements class for generic process profiles.* Requirement 23-3 *Requirements class for process implementation profiles.* Requirement 23-4 *Requirements class for process profile inheritance.*

8.5.1. Process Concept

A process concept is an object that provides high-level documentation about a general group of processes. It describes the purpose, methodology and properties of a process but not the specific input and output parameters. It is rather a documentation resource that may be referenced by refined process definitions to document their relation to a common principle.

Example: The concept "Buffer" may be used to describe all Buffer operations. More specific Buffer processes may be defined on raster or vector data models, be performed on a geoid or in CRS units, have further inputs, such as distance or tolerance and may even perform additional computations such as dissolve or line cap styling.

Due to the heterogeneity of process definitions and the variety of documentation requirements, there is no general information model for process concepts. Most of the time, concepts will be documented in HTML or similar multimedia formats. Formally, a concept consists of a unique identifier and a descriptive document.

REQUIREMENTS CLASS 24	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process
NORMATIVE STATEMENTS	Requirement 24-1 *Process concepts are documentation resources with identity. They shall have an identifier that may be referenced by more detailed process definitions.* Requirement 24-2 *The content of a process concept resource shall eventually lead to the definition of computing processes.*

8.5.2. Generic Process Profile

A generic profile is the abstract interface of a process. It provides a detailed description of the process mechanics and declares a signature for process inputs and outputs. The generic profile consists of a unique identifier and a generalized process description. This is similar to a process description as defined in Clause 8.4 but does not provide a definition of supported data exchange formats.

Example: A generic profile for a Buffer operation may be derived from the Buffer definition in the ISO standard for simple feature access [ISO 19125-1:2006]. The buffer method on simple features "Returns a geometric object that represents all points whose distance from this geometric object is less than or equal to distance. Calculations are in the spatial reference system of this geometric object" (ISO 19125-1:2006, subclause 6.1.2.4). This definition is specific about the conceptual data model details the behavior of the buffer method and the treatment of CRS units. In addition, it defines the name and data type of the distance parameter.

This generic definition of a Simple features buffer process may be inherited by multiple implementations that use arbitrary encodings for input and output data. The important part here is the well-defined behavior of the process at a generic level and the standardization of parameter names.[6]

REQUIREMENTS CLASS 25	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model/process http://www.opengis.net/spec/WPS/2.0/req/native-process/model/description-type http://www.opengis.net/spec/WPS/2.0/req/native-process/model/description IETF RFC 4646
NORMATIVE STATEMENTS	Requirement 25-1 *A process description shall comply with the structure defined in Figure 13 and Table 18.* Requirement 25-2 *The language of the human-readable elements within the process description shall be identified by a language identifier as specified in IETF RFC 4646.* Requirement 25-3 *The description of process inputs and outputs shall comply with the structure defined in Figure 13, Table 19, and Table 20.*

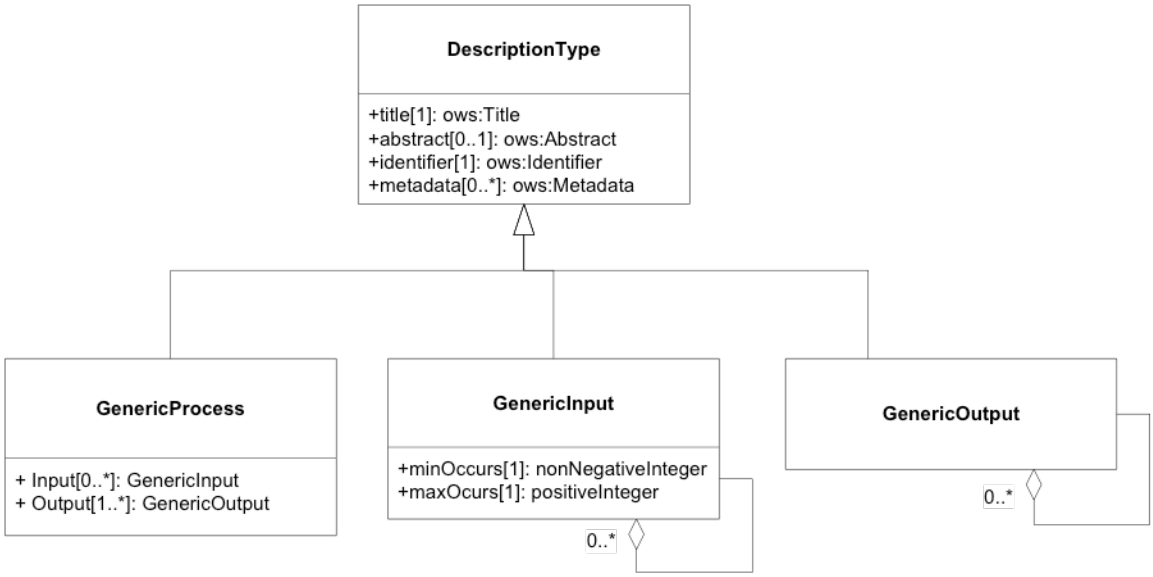

Figure 13 — GenericProcess UML class diagram

[6]NOTE: In terms of names, the Buffer definition in [ISO 19125-1:2006] only covers the distance parameter since it assumes a Geometry object providing that method. A complete generic profile would also have to define names for input and output geometries.

Table 18 — The GenericProcess structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Title			
Abstract			
Keywords	Inherited from Table 4		
Identifier			
Metadata			
Language	Language identifier for the human readable process description elements.	Character String. This language identifier shall be as specified in IETF RFC 4646.	One (mandatory)
Input	Input items (arguments) of a process.	GenericInput structure, see Table 19.	Zero or more (optional)
Output	Output items (results) of a process	GenericOutput structure, see Table 20.	One or more (mandatory)

Table 19 — Parts of the GenericInput structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Title			
Abstract			
Keywords	Inherited from Table 4		
Identifier			
Metadata			
min Occurs[a]	Minimum number of times that values for this parameter are required	Non-negative integer; defaults to "1", '0' means the input is optional.	Zero or one (optional)
max Occurs[a]	Maximum number of times that this parameter may be present	Non-negative integer, defaults to "1".	Zero or one (optional)
Input	Nested Input.[b]	GenericInput structure, Table 19 (this table).	Zero or more (optional)

[a] The minOccurs and maxOccurs parameters have identical semantics to the like-named XML Schema occurrence constraints.

[b] It is recommended to keep the nesting level as low as possible.

Table 20 — Parts of the GenericOutput structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Title			
Abstract			
Keywords	Inherited from Table 4		
Identifier			
Metadata			
Output	Nested Output.[a]	GenericOutput structure, Table 20 (this table).	Zero or more (conditional)[a]

[a] It is recommended to keep the nesting level as low as possible.

8.5.3. Process Implementation Profile

Implementation profiles cover all descriptive elements of a process down to the supported data exchange formats. Technically they are process descriptions, but with the scope of a process profile, i.e., a harmonized and well-defined computing process that may be implemented by multiple service providers.

REQUIREMENTS CLASS 26	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/native-process/model/process-description
NORMATIVE STATEMENT	Requirement 26-1 _A process implementation profile description shall comply with the process description structure defined in http://www.opengis.net/spec/WPS/2.0/req/native-process/model/process-description._

8.5.4. Profile Inheritance

The hierarchical structure allows for inheritance between different types of profiles (see Figure 14). The definition and use of generic profiles for commonly used processing functions is generally recommended. However, there might be use cases for a product-specific harmonization of process interfaces. In this case, an implementation profile may be directly derived from a concept or defined in isolation.

If a profile in the hierarchy is derived from another profile at an equal or higher level, it must respect the inheritance rules given in Table 21. These rules ensure consistency in the use of identifiers, the specification of inputs and outputs as well as compliance to the behavior of the declared parent profiles.

Processes and process profiles may use metadata links to indicate compliance to a particular process profile. Links to particular profiles shall be embedded in the process' metadata elements. The reserved role identifiers for the different profile levels are listed in Table 22.

An example which illustrates the inheritance rules for process profiles is given in Annex B.10.

REQUIREMENTS CLASS 27	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/native-process/model/profile/concept http://www.opengis.net/spec/WPS/2.0/req/native-process/model/profile/generic http://www.opengis.net/spec/WPS/2.0/req/native-process/model/profile/implementation http://www.opengis.net/spec/WPS/2.0/req/native-process/model/description
NORMATIVE STATEMENTS	Requirement 27-1 *Process profiles may inherit properties from profiles specified at a higher or equal level. The hierarchy is defined in Figure 14.* Requirement 27-2 *Inheriting process profiles and processes shall obey to the inheritance rules defined in Table 21.* Requirement 27-3 *Links to particular profiles are expressed in the DescriptionType's metadata element as specified in Table 5.* Requirement 27-4 *The links target for profile metadata is the URL of the process profile.* Requirement 27-5 *The link roles defined in Table 22 shall be used to express the hierarchical level of the linked profile.*

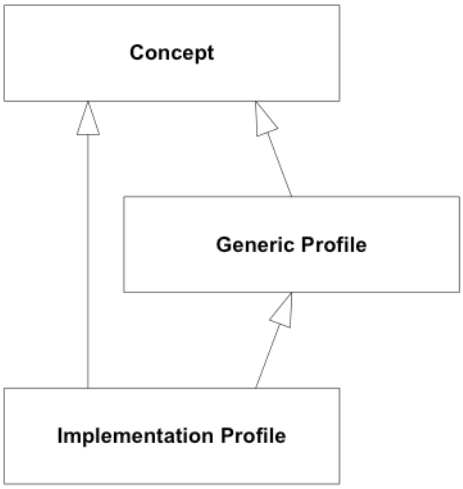

Figure 14 — Inheritance hierarchy for process profiles UML class diagram

Table 21 — Inheritance and override rules for process profiles

PROPERTY	GENERIC PROFILE	IMPLEMENTATION PROFILE	IMPLEMENTATION (INSTANCE LEVEL)
Process			
Identifier	D	O	O
Title	D	O	O
Keywords	D	E	E
Abstract	D	O	O
Metadata	D	E/O[a]	E/O[a]
Input			E[b]
Identifier	D	I	I
Title	D	I	I
Keywords	D	E	E
Abstract	D	I	I
Metadata	D	E/O[a]	E/O[a]

PROPERTY	GENERIC PROFILE	IMPLEMENTATION PROFILE	IMPLEMENTATION (INSTANCE LEVEL)
Multiplicity	D	R[c]	E[d]
Data format		D	E[d]
Output			E[b]
Identifier	D	I	I
Title	D	I	I
Keywords	D	E	E
Abstract	D	I	I
Metadata	D	O	O
Data format		D	E[d]

NOTE:

D	Declare (Introduction of a new property)
I	Inherit (A property is inherited from a superior level AND its value remains unchanged.)
O	Override (A property is inherited from a superior level AND its value is overridden.)
E	Extend (A property is inherited from a superior level AND its value is extended.)
R	Restrict (A property is inherited from a superior level AND its value is restricted.)

[a] The list of metadata references to superior process profiles shall be extended. Documentation metadata may be overridden, i.e. replaced by other documentation resources.

[b] Additional optional inputs or supplementary outputs may be added here.

[c] Implementation profiles may restrict the maximum cardinality of a superior generic profile (e.g. for theoretically infinite inputs). They shall not modify the minimum cardinality.

[d] Implementations may allow more or larger input datasets than the implementation profile, or support additional data exchange formats.

Table 22 — Role identifiers for process profiles

PROFILE LEVEL	URI
Process concept	http://www.opengis.net/spec/wps/2.0/def/process-profile/concept
Generic process profile	http://www.opengis.net/spec/wps/2.0/def/process-profile/generic
Process Implementation profile	http://www.opengis.net/spec/wps/2.0/def/process-profile/implementation

9

WPS NATIVE PROCESS MODEL ENCODING

9 WPS NATIVE PROCESS MODEL ENCODING

9.1. XML Schema Implementation

This section specifies the XML encoding of the elements of the WPS native process model. The referred XML schema elements are provided by the associated schema package delivered with this standard and located at http://schemas.opengis.net/wps/2.0/.

REQUIREMENTS CLASS 28	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/native-process/model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 28-1 *Requirements class for data type XML encoding.* Requirement 28-2 *Requirements class for process description XML encoding.* Requirement 28-3 *Requirements class for generic profile XML encoding.*

9.1.1. Data Types

The XML encoding of data types defines encoding rules for ComplexData, LiteralData, BoundingBoxData as well as their values.

Examples for data type encodings are listed in Annex B.1.

REQUIREMENTS CLASS 29	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/native-process/model/datatypes/complex-data http://www.opengis.net/spec/WPS/2.0/req/native-process/model/datatypes/literal-data http://www.opengis.net/spec/WPS/2.0/req/native-process/model/datatypes/bounding-box-data OWS Common 2.0

REQUIREMENTS CLASS 29	
NORMATIVE STATEMENTS	Requirement 29-1 *The XML encoding of ComplexData, LiteralData, BoundingBoxData and their values shall comply with the XML schema provided by dataTypes.xsd.* Requirement 29-2 *The mime type for XML encoded literal and bounding box data values shall be "text/xml".*

9.1.2. Process Description

This clause specifies the XML encoding for the Process description.

A Process example is listed in Annex B.2.

REQUIREMENTS CLASS 30	
OBLIGATION	requirement
TARGET TYPE	Derived software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/native-process/model/description OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 30-1 *An XML encoded Process description shall be a valid XML document of the type wps:Process.* Requirement 30-2 _The content of the XML encoded Process description shall comply with the content of the information elements defined by the requirements class http://www.opengis.net/spec/WPS/2.0/req/native-process/model/description._

9.1.3. Generic Process

This clause specifies the XML encoding for the GenericProcess.

A GenericProcess example is listed in Annex B.3.

REQUIREMENTS CLASS 31	
OBLIGATION	requirement
TARGET TYPE	Derived software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/native-process/profile/generic OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 31-1 *An XML encoded description of a Generic Process shall be a valid XML document of the type wps:GenericProcess.*

REQUIREMENTS CLASS 31	
	Requirement 31-2 _The content of the XML encoded description of a Generic Process shall comply with the content of the information elements defined by the requirements class http://www.opengis.net/spec/WPS/2.0/req/native-process/model/profile/generic._

9.2. Plain Text Encoding for LiteralData and BoundingBoxData Values

This clause specifies the plain text encoding of data types for literal and bounding box data values.

REQUIREMENTS CLASS 32	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/native-process/model/datatypes/complex-data http://www.opengis.net/spec/WPS/2.0/req/native-process/model/datatypes/literal-data http://www.opengis.net/spec/WPS/2.0/req/native-process/model/datatypes/bounding-box-data OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 32-1 *The plain text encoding of literal and bounding box values comply with the BNF schema below.* Requirement 32-2 *The mime type for plain text encoded literal and bounding box values shall be "text/plain".*

```
Literal values - BNF schema:
literalvalue   = value *1("@datatype=" datatype) *1("@uom=" uom)
value          = 1*VCHAR
datatype  = URI
uom       = URI

Literal values - Example:
70@datatype=http://www.w3.org/2001/XMLSchema#integer@uom=meter

BoundingBox values - BNF schema[7]:
```

[7]The dimensions attribute is included for compliance with the BoundingBox structure defined in OWS Common [OGC 06-121r9]. However, its use should be

```
bbox      = lc_coords "," uc_coords ["," crs] ["," dimensions]
lc_coords = number ["," number]
uc_coords = number ["," number]
number         = 1*DIGIT["." 1*DIGIT]
crs       = 1*VCHAR
dimensions = 1*DIGIT

BoundingBoxData values - Examples:
51.9,7.0,53.0,8.0,EPSG:4326

51.9,7.0,53.0,8.0,http://www.opengis.net/def/crs/EPSG/0/4258
```

Listing

generally avoided since the number of dimensions is already part of the CRS definition and usually superfluous.

10

COMMON WPS SERVICE MODEL

10 COMMON WPS SERVICE MODEL

A Web Processing Service consists of processes and service operations. By definition, processes represent the computational functionality of a WPS, while service operations are used to interact with the WPS and in particular to use the service's process offerings.

REQUIREMENTS CLASS 33	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENT	Requirement 33-1 *Requirements class for data transmission.*
DESCRIPTION	*Requirements class for WPS service handling.*
Example:	
NORMATIVE STATEMENTS	Requirement 33-2 *Requirements class for common process offering properties.* Requirement 33-3 *Requirements class for the status information document.* Requirement 33-4 *Requirements class for the processing result document.* Requirement 33-5 *Requirements class for the GetCapabilities operation.* Requirement 33-6 *Requirements class for the DescribeProcess operation.* Requirement 33-7 *Requirements class for the Execute operation.* Requirement 33-8 *Requirements class for the GetStatus operation.* Requirement 33-9 *Requirements class for the GetResult operation.*

10.1. Overview of WPS Core Operations

The WPS interface specified in this section supports retrieval and execution of processes for geospatial computation. For that purpose, the WPS service model specifies the following operations that may be invoked by a WPS client and performed by a WPS server[8]:

GetCapabilities — This operation allows a client to request information about the server's capabilities and processes offered (see Clause 10.7).

[8]NOTE: Future extensions of this specification may introduce additional operations.

DescribeProcess — This operation allows a client to request detailed metadata on selected processes offered by a server (see Clause 10.8).

Execute — This operation allows a client to execute a process comprised of a process identifier, the desired data inputs and the desired output formats (see Clause 10.9).

GetStatus — This operation allows a client to query status information of a processing job (conditional, see Clause 10.10).

GetResult — This operation allows a client to query the results of a processing job (conditional, see Clause 10.11).

During a sequence of WPS requests, a client should first issue a GetCapabilities request to the server to obtain an up-to date listing of available processes. Then, it may issue a DescribeProcess request to find out more details about particular processes offered, including the supported data formats. To run a process with the desired input data, a client will issue an Execute request[9] (Figure 15).

The operations GetStatus and GetResult are used in conjunction with asynchronous execution. If a WPS server offers synchronous process execution only, these operations may not be implemented. Detailed guidance is provided by the corresponding profiles (Clause 10.12 and Clause 10.13) and conformance classes.

[9]NOTE: A WPS server can change its offering at any time, in particular between a GetCapabilities, a subsequent DescribeProcess, and a subsequent Execute request. Any quality of service (QoS) guarantees are within the responsibilities of the service provider and not covered by this standard.

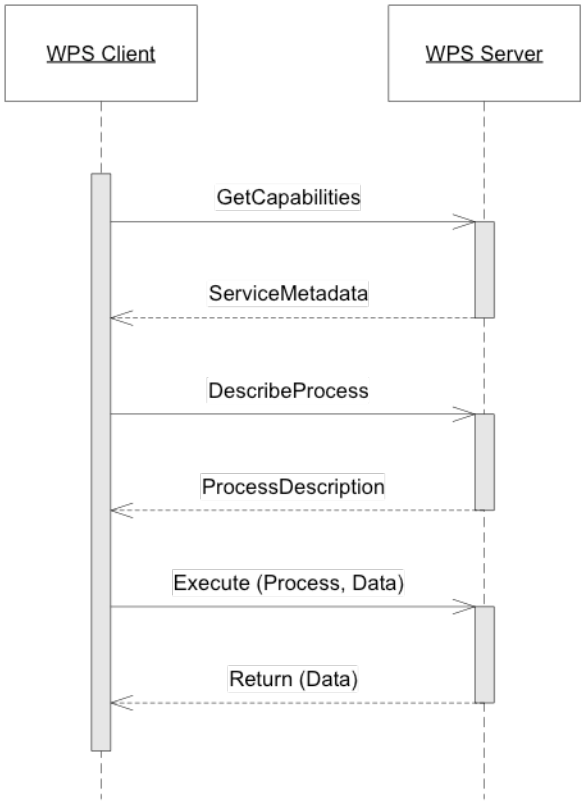

Figure 15 — Common sequence of WPS operations UML sequence diagram

10.2. Data Transmission

Data exchange between WPS clients and servers requires an agreement on the general data exchange patterns and suitable communication protocols. Data may be sent to (received from) a WPS in two distinct ways: (1) by reference (using HTTP/GET or HTTP/POST), and (2) by value. Clients may send input data in either fashion. Output data may be requested in any fashion declared by the data transmission options defined for the process offering. Typically, large data inputs and outputs are delivered by reference.

REQUIREMENTS CLASS 34	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model
NORMATIVE STATEMENT	Requirement 34-1 *The data transmission structures for process inputs and outputs shall comply with the structures defined in Figure 16, Table 23, Table 24, Table 25, Table 26, and Table 27.*

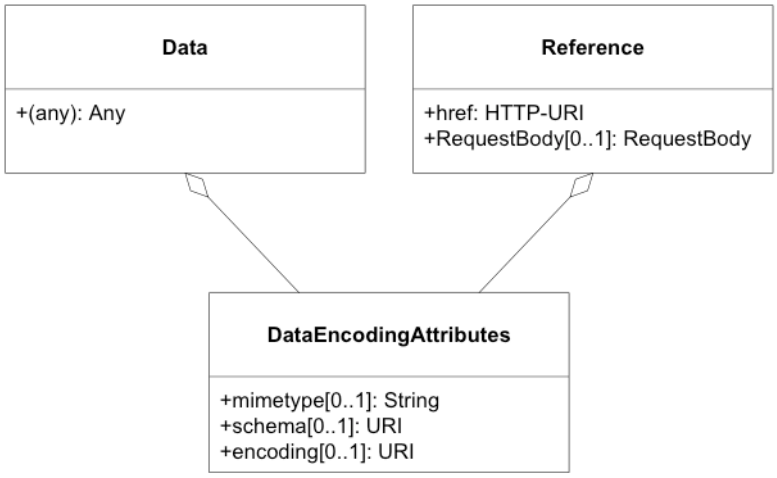

Figure 16 — Input and output data transmission structures UML class diagram

Table 23 — Parts of the inline Data structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
mimetype			
encoding	See Table 24 — Properties of the DataEncodingAttributes structure.		
schema			
(any)[a]	The actual input or output data.	Any type and value	One (mandatory)

[a] The data is embedded here as part of the Data element, in the mimeType, encoding, and schema indicated by the first three parameters if they exist, or by the relevant defaults.

Table 24 — Properties of the DataEncodingAttributes structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
mimetype	MimeType of the data.	Character String	One (mandatory)
encoding	Well-known encoding or character set of the data.	URI “raw” shall be used for plain binary data “base64” shall be used for base64 encoded data Character set identifiers (e.g. “UTF-8”) shall be used for text or CSV data.	Zero or one (conditional)[a]
schema	Identification of the data schema.	URI	Zero or one (conditional)[a]

[a] This shall be provided if: 1) the process data item supports multiple encodings / schemas, and 2) the data is not of the default encoding / schema, and 3a) the schema / encoding cannot be retrieved from the data itself, or 3b) the

encoding / schema information is deeply buried inside the data (i.e. not part of some header) and requires significant parsing effort.

Table 25 — Parts of the Reference structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
mimetype			
encoding	See Table 24 — Properties of the DataEncodingAttributes structure.		
schema			
href	HTTP URI that points to the remote resource where the data may be retrieved.	HTTP URI	One (mandatory)
Request Body	Request body element that is used for HTTP/POST requests to the above URL. If no request body is present, an HTTP/GET	Request should be used to retrieve the data. RequestBody structure, see Table 26.	Zero or one (optional)

Table 26 — Parts of the RequestBody structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Body	The contents of this element to be used as the body of the HTTP request message to be sent to the service identified in ../Reference/@href. For example, it could be an XML encoded WFS request using HTTP/POST.	Any type	Zero or one (conditional)[a]
BodyReference	Reference to a remote document to be used as the body of an HTTP/POST request message to the service identified in the href element in the Reference structure (Table 25).	BodyReference, see Table 27.	Zero or one (conditional)[a]

[a] One and only one of these items shall be included.

Table 27 — Parts of the BodyReference structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
href	HTTP URI that points to the remote resource where the request body may be retrieved.	HTTP URI	One (mandatory)

10.3. WPS Service Handling

The WPS service model seeks compliance with OWS Common and implements the baseline communication protocol for OWS services and the related information elements defined in [OGC 06-121r9], clause 9.2.1.

REQUIREMENTS CLASS 35	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model
NORMATIVE STATEMENTS	Requirement 35-1 *All request types of a WPS, except GetCapabilities, shall use the minimum request parameters, defined in [OGC 06-121r9].* Requirement 35-2 *For all WPS request types, the service parameter shall have a fixed value of "WPS".* Requirement 35-3 *For all WPS request types, the request version parameter shall have a fixed value of "2.0.0".* Requirement 35-4 *All request types of a WPS, except GetCapabilities, shall comply with the request type structure defined in Table 28.*

Table 28 — Properties of the RequestBaseType

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
service	Service type identifier	Character String, fixed to "WPS"	One (mandatory)
version	Specification version for operation	Character String, fixed to "2.0.0"	One or more (mandatory)
Extension	Any ancillary information to be sent from client to server. Placeholder for further request parameters defined by WPS extension standards.	Any type	Zero or more (optional)

10.4. Process Offering

A process offering structure contains information about the processes that may be run on a WPS server. Furthermore, the process offerings structure contains properties that describe the

available execution modes of a process, the allowed data transmission modes, its version, and the process type (if it deviates from the native process model).

REQUIREMENTS CLASS 36	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model
NORMATIVE STATEMENT	Requirement 36-1 *For all process offerings, the attributes defined in Table 29 shall be specified.*

Table 29 — Parts of the ProcessOfferingPropertiesAttributes structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
jobControlOptions	Job control options supported for this process	List of supported options for process control (see Table 30), extensions may introduce additional control options.	One or more (mandatory)
output Transmission	Supported transmission modes for output data (by value / by reference)	List of supported data transmission options (see Table 31).	One or more (mandatory)
processVersion	Release version of process (not of WPS specification). May be specified to reflect updates or changes in the process offering.	ows:VersionType	Zero or one (optional) Include when needed to identify process version.[a]
processModel	Type of the process description	HTTP-URI. Value is defined by the process description specification. Defaults to "native".	Zero or one (conditional) Include when using a different process model than the native process model.[b]

[a] The processVersion is informative only. Version negotiation for processVersion is not available. Requests to Execute a process do not include a processVersion identifier.

[b] This is an extension hook to support processes that have been specified in other OGC Standards, such as SensorML. For those process models, compliance with the WPS abstract process model (Clause 7.3) has to be ensured.

Table 30 — Basic job control options

OPTION	DEFINITION
sync-execute	The process offering can/shall be executed synchronously.
async-execute	The process offering can/shall be executed asynchronously.

Table 31 — Data transmission options

OPTION	DEFINITION
value	The data is delivered by value.
reference	The data is delivered by reference.

10.5. StatusInfo Document

The StatusInfo document is used to provide identification and status information about jobs on a WPS server.

REQUIREMENTS CLASS 37	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model
NORMATIVE STATEMENTS	Requirement 37-1 *The StatusInfo document shall comply with the structure defined in Table 32.* Requirement 37-2 *The Status element shall contain a well-known identifier for the JobStatus. A basic set has been defined in the WPS conceptual model. WPS operations and WPS extensions may define additional states.* Requirement 37-3 *The content of the Status element shall be treated case insensitive.*

Table 32 — StatusInfo structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
JobID	Unambiguously identifier of a job within a WPS instance.	Character String[a]	One (mandatory)
Status	Well-known identifier describing the status of the job.	Character String[b]	One (mandatory)
ExpirationDate	Date and time by which the job and its results will be no longer accessible.[c]	ISO-8601 date/time string in the form YYYY-MM-DDTHH:MM:SS.SSSZ with T separator character and Z suffix for coordinated universal time (UTC)	Zero or one (optional) Include if required.
Estimated Completion	Date and time by which the processing job will be finished.	ISO-8601 date/time string in the form YYYY-MM-DDTHH:MM:SS.SSSZ with T separator character and Z suffix for coordinated universal time (UTC)	Zero or one (optional) Include if available.
NextPoll	Date and time for the next suggested status polling.	ISO-8601 date/time string in the form YYYY-MM-DDTHH:MM:SS.SSSZ with T separator character and Z suffix for coordinated universal time (UTC)	Zero or one (optional) Include if required.
PercentCompleted	Percentage of process that has been completed.	Integer{0..100}[d]	Zero or one (optional) Include if available.

[a] Particularly suitable JobIDs are UUIDs or monotonic identifiers such as unique timestamps. If the privacy of a Processing Job is imperative, the JobID should be non-guessable.

[b] The basic status set is defined in Table 3. Additional states may be defined by certain operations or extensions of this standard.

[c] This element will usually become available when the execution has finished (Status = “finished”).

[d] Zero (0) means the execution has just started, and 100 means the job is complete. This value is informative only without any accuracy guarantees.

10.6. Result Document

A Result document is a structure that contains the results of a process execution. It is a shared element between the Execute and GetResult operations.

REQUIREMENTS CLASS 38	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model http://www.opengis.net/spec/WPS/2.0/req/service/model/data-transmission
NORMATIVE STATEMENT	Requirement 38-1 *The Result document shall comply with the structure defined in Table 33.*

Table 33 — Result structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
JobID	Unambiguously identifier of a job within a WPS instance.	Character String[a]	Zero or one (conditional)[b]
Expiration Date	Date and time by which the results will be no longer accessible.[c]	ISO-8601 date/time string in the form YYYY-MM-DDTHH:MM:SS.SSSZ with T separator character and Z suffix for coordinated universal time (UTC)	Zero or one (conditional) Include if required, i.e. if the server will delete stored results at some point in time.
Output	Output item returned by a process execution.	DataOutputType structure, see Table 37.	One or more (mandatory)

[a] Particularly suitable JobIDs are UUIDs or monotonic identifiers such as unique timestamps. If the privacy of a Processing Job is imperative, the JobID should be non-guessable. In asynchronous execution, the JobID would be shared among related StatusInfo and Result documents.

[b] Include if required, e.g. in a response to an asynchronous execution.

[c] For results delivered "by reference" this element may indicate when the Data Outputs will be deleted by the server.

Table 34 — Parts of the DataOutputType structure

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
id	Unambiguous identifier or name of an output item.	URI	One (mandatory)
Data	The data provided by this output item.		Zero or one (conditional)[a]

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Output	Nested output, child element.	DataOutputType structure, see Table 34 (this table)	Zero or one (conditional)[a]

[a] One and only one of these items shall be included.

10.7. GetCapabilities Operation

Per OGC 06-121r9, a GetCapabilities operation is required for any OGC Web service. For WPS, this operation allows a client to retrieve service metadata, basic process offerings, and the available processes present on a WPS server.

REQUIREMENTS CLASS 39	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 39-1 *Requirements class for the GetCapabilities request.* Requirement 39-2 *Requirements class for the GetCapabilities response.* Requirement 39-3 *Requirements class for GetCapabilities operation exceptions.*

10.7.1. GetCapabilities Request

The GetCapabilities request is mandatory for any OGC service. Figure 17 shows how the GetCapabilities request for a WPS relates to the generic GetCapabilitiesType defined by OWS Common. An Extension element provides a hook for further request parameters that may be defined by WPS extension specifications.

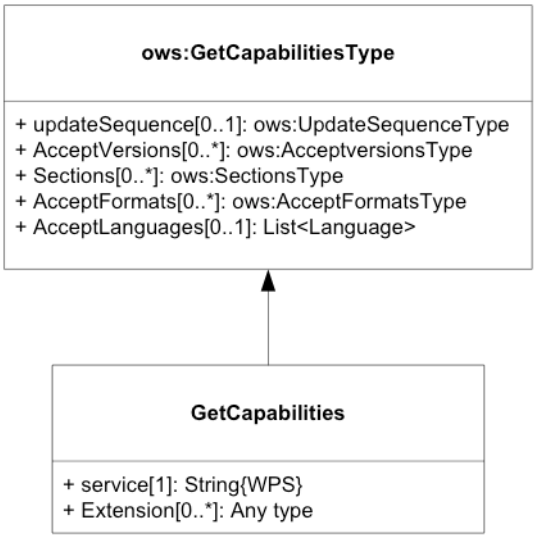

Figure 17 — GetCapabilities request UML class diagram

REQUIREMENTS CLASS 40	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 40-1 *The GetCapabilities operation request shall be implemented as specified in Clause 7 of OWS Common [OGC 06-121r9].* Requirement 40-2 *If the AcceptVersions parameter is contained in the request, it shall contain the character string "2.0.0".* Requirement 40-3 *In addition to the properties inherited from OWS Common GetCapabilities, the WPS GetCapabilities request shall include the properties according to Table 35.*

Table 35 — Additional properties in the GetCapabilities request

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Service	Service type identifier	Character string, fixed to "WPS"	One (mandatory)
Extension	Container for elements defined by extension specifications	Any type. Value is defined by the extension specification.	Zero or more (optional)

10.7.2. GetCapabilities Response

The response to a GetCapabilities operation is a document describing the service's capabilities. Figure 18 shows how the WPS Capabilities are derived from the CapabilitiesBaseType defined in [OGC 06-121r9]. The OperationsMetadata element lists the request types supported by a WPS server. The contents section delivers information about the process offerings of the server. An Extension element provides a hook for additional service capabilities that cannot be covered by other available elements.

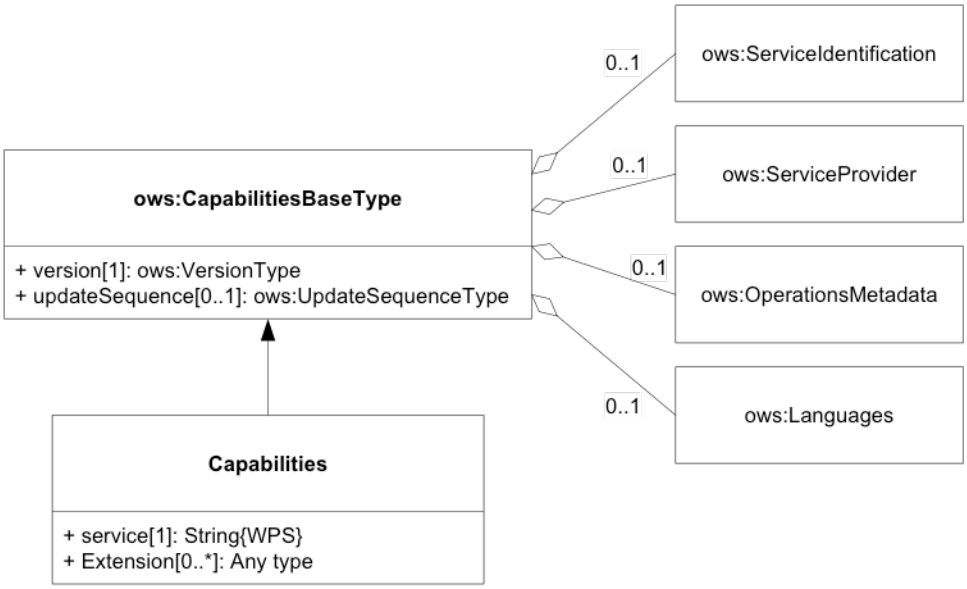

Figure 18 — Capabilities document UML class diagram

REQUIREMENTS CLASS 41	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 41-1 *The capabilities response shall provide service metadata according to [OGC 06-121r9], clause 7.4.2.* Requirement 41-2 *The default version of the Capabilities document returned by a service implementing this standard shall be "2.0.0".* Requirement 41-3 *In addition to the properties inherited from OWS Common OWSService Metadata, the WPS Capabilities shall include the properties according to Table 36.* Requirement 41-4 *The contents section of the Capabilities document shall contain a process summary for each of the process offerings.*

REQUIREMENTS CLASS 41
Requirement 41-5 *The properties of a process summary shall provide the properties defined in Table 37.*

Table 36 — Additional properties in the Capabilities document

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
service	Service type identifier	Character string, fixed to "WPS"	One (mandatory)
Contents	List of brief descriptions of the processes offered by this WPS server.	ProcessSummary, see Table 37	One (mandatory)
Extension	container for elements defined by extension specifications	Any type. Value is defined by the extension specification.	Zero or more (optional)

Table 37 — Properties of the ProcessSummary

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Title	Title of a process, normally available for display to a human.	ows:Title	One (mandatory)
Abstract	Brief narrative description of a process, normally available for display to a human.	ows:Abstract	Zero or more (optional)
Keywords	Keywords that characterize a process,	ows:Keyword	Zero or more (optional)
Identifier	Unambiguous identifier or name of a process.	ows:Identifier[a]	One (mandatory)
Metadata	Reference to more metadata about this item.	ows:Metadata	Zero or more (optional) Include when available and useful
processModel	Inherited from Table 29.		
jobControlOptions			
output Transmission			

[a] Additional content such as separate code space and version attributes in the Identifier element are not allowed.

10.7.3. GetCapabilities Exceptions

If a WPS server encounters an error while performing a GetCapabilities operation, it shall return an exception report as specified in Clause 7.4 of [OGC 06-121r9].

REQUIREMENTS CLASS 42	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENT	Requirement 42-1 *If a WPS server encounters an error while performing a GetCapabilities operation, it shall return an exception report message as specified in subclause 7.4 of [OGC 06-121r9].*

10.8. DescribeProcess Operation

The DescribeProcess operation allows WPS clients to query detailed process descriptions for the process offerings.

REQUIREMENTS CLASS 43	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 43-1 *Requirements class for the DescribeProcess request.* Requirement 43-2 *Requirements class for the DescribeProcess response.* Requirement 43-3 *Requirements class for DescribeProcess operation exceptions.*

10.8.1. DescribeProcess Request

The DescribeProcess request inherits basic properties from the RequestBaseType. An Identifier element shall contain a list of the process identifiers for which the process descriptions shall be

obtained. If the service supports multilingual process descriptions, the desired language of the free-text elements in the process description may be queried with a language parameter.

REQUIREMENTS CLASS 44	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0 IETF RFC 4646 http://www.opengis.net/spec/WPS/2.0/req/service/model/handling
NORMATIVE STATEMENTS	Requirement 44-1 *The DescribeProcess request shall comply with the structure defined in Figure 19 and Table 38.* Requirement 44-2 *The reserved character string "ALL" shall be used to obtain a response that contains descriptions of all process offerings provided by the server.*

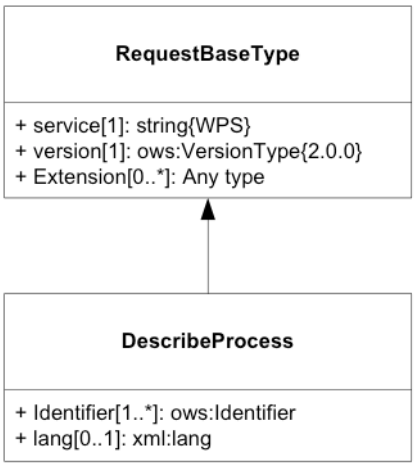

Figure 19 — DescribeProcess request UML class diagram

Table 38 — Additional properties in the DescribeProcess request

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Identifier	One or more process identifiers for which the detailed description shall be obtained.	ows:Identifier Value shall be one of the process identifiers listed in the ProcessSummary elements in the Capabilities document. The fixed value "ALL" indicates that the description of all process offerings shall be returned.	One or more (mandatory)
lang	Desired language of the process description.	xml:lang IETF RFC 4646 language code of the human-readable text elements in the process description (e.g. "en"). This shall be one of the languages defined in the Capabilities document.	Zero or one (optional)

10.8.2. DescribeProcess Response

The response to a DescribeProcess operation is a ProcessOfferings document. This document contains a ProcessOfferings section for each available process on the server. In contrast to the ProcessSummary in the server's capabilities, the processes are described in their declared description format.

REQUIREMENTS CLASS 45	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model IETF RFC 4646
NORMATIVE STATEMENTS	Requirement 45-1 *The DescribeProcess response shall comply with the structure defined in Table 39 and Table 40.* Requirement 45-2 *If the client has requested a particular response language, the process descriptions shall be delivered in that language.*

Table 39 — Properties in the ProcessOfferings document

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
lang	Language in which the process offerings are described.	xml:lang IETF RFC 4646 language code of the human-readable text elements in the process description (e.g. "en"). This shall be the language identified in the DescribeProcess request.	Zero or one (optional)
ProcessOfferings	List of ProcessOfferings.	ProcessOffering, defined in Table 40.	One or more (optional)

Table 40 — ProcessOffering properties

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
processModel			
jobControlOptions	Inherited from Table 29		
output Transmission			

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Process	Native Process description.	Process type, as defined in the native process model.	Zero or one (conditional)[a]
Any	Any other well-defined process description, identified by the processType.	Any type. Must conform to requirements associated with the declared processType.	Zero or one (conditional)[a]

[a] One and only one of these items shall be included.

10.8.3. DescribeProcess Exceptions

If a WPS server encounters an error while performing a DescribeProcess operation, it shall return an exception report as specified in Clause 8 of [OGC 06-121r9]. If the error was encountered due to an invalid process identifier, the server shall respond with the exception code defined in Table 41.

REQUIREMENTS CLASS 46	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 46-1 *If a WPS server encounters an error while performing a DescribeProcess operation, it shall return an exception report message as specified in Clause 8 of [OGC 06-121r9].* Requirement 46-2 *If the error was encountered due to an invalid process identifier, the server shall respond with the exception code defined in Table 41.*

Table 41 — Additional exception codes for the DescribeProcess operation

EXCEPTIONCODE VALUE	EXCEPTIONTEXT	LOCATOR	HTTP STATUS CODE
NoSuchProcess	One of the identifiers passed does not match with any of the processes offered by this server.	List of violating process identifiers.	400 (Bad request)

10.9. Execute Operation

The Execute operation allows WPS clients to run a specified process implemented by a server, using the input parameter values provided and returning the output values produced. Inputs may be included directly in the Execute request (by value), or reference web accessible resources (by reference). The outputs may be returned in the form of an XML response document, either embedded within the response document or stored as web accessible resources. Alternatively, for a single output, the server may be directed to return that output in its raw form without being wrapped in an XML response document. This is illustrated in Figure 20.

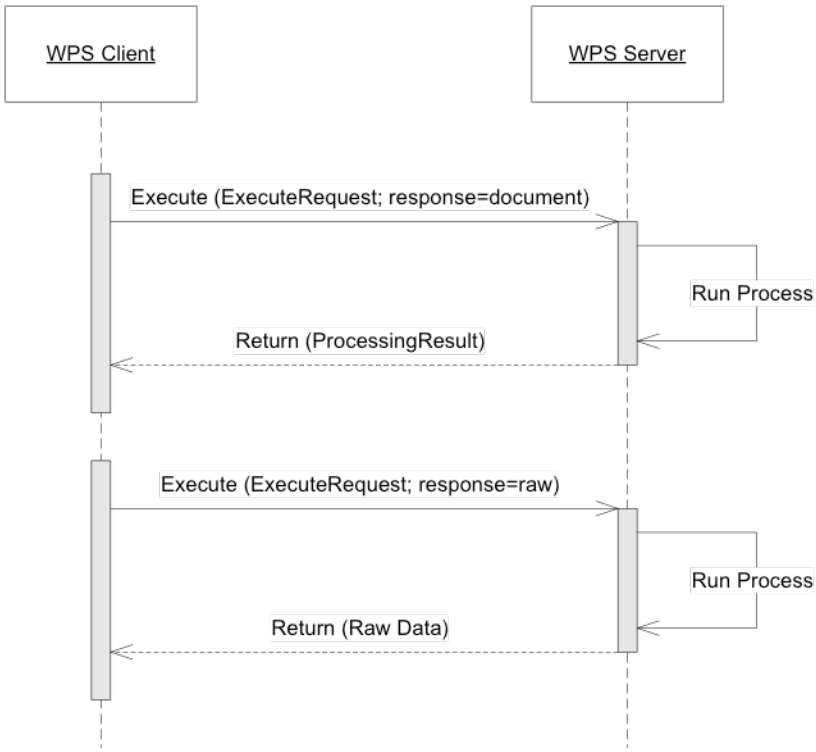

Figure 20 — Execute response document and raw data UML sequence diagram

REQUIREMENTS CLASS 47	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 47-1 *Requirements class for the Execute request.* Requirement 47-2 *Requirements class for the Execute response.* Requirement 47-3 *Requirements class for Execute operation exceptions.*

10.9.1. Execute Request

The Execute request is a common structure for synchronous and asynchronous execution. It inherits basic properties from the RequestBaseType and contains additional elements that identify the process that shall be executed, the data inputs and outputs, and the response type of the service.

REQUIREMENTS CLASS 48	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0 http://www.opengis.net/spec/WPS/2.0/req/service/model/handling
NORMATIVE STATEMENTS	Requirement 48-1 *The Execute request shall comply with the structure defined in Figure 21, Table 42, Table 43, and Table 44.* Requirement 48-2 *Raw data output shall only be requested if the processes execution will deliver one single output value. (For a process that provides more than one output, the desired output has to be specified in the request for raw data delivery.)*

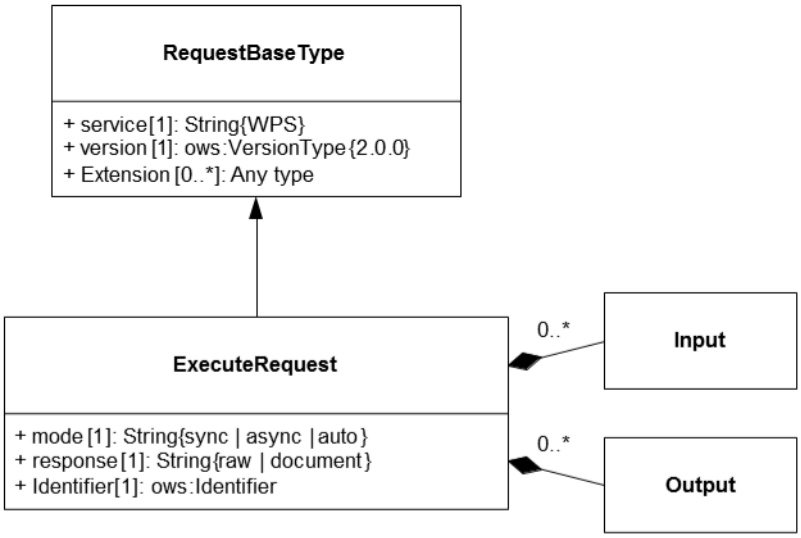

Figure 21 — Execute request UML class diagram

Table 42 — Additional properties in the Execute request

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
response	Desired response format, i.e. a response document or raw data.	String {raw[a] | document}	One (mandatory)
mode	Desired execution mode.	String{sync | async | auto[b]} Valid values are to be derived from the jobControlOptions property of each ProcessOffering. "auto" delegates the choice of execution mode to the server.	One (mandatory)
Identifier	Unambiguous identifier of the process that shall be executed.	ows:Identifier Value shall be one of the process identifiers listed in the ProcessSummary elements in the Capabilities document.	One (mandatory)
Input	Data inputs provided to this process execution.	DataInputType structure, see Table 43.	Zero or more (conditional)[c]
Output	Specification of outputs expected from the process execution, including the desired format and transmission mode for each output.	OutputDefinitionType structure, see Table 44.	Zero or more (conditional)[d]

[a] Raw output shall only be requested if the execution would return a single output.

[b] In the case of auto, the server shall respond quickly to avoid connection timeouts.

[c] If a process does not have any inputs, or if, for all required inputs the process description contains default values, this element may be omitted.

[d] If a process shall return all outputs in their default format, this element may be omitted.

Table 43 — Properties of the DataInputType

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
id	Identifier of a particular input, as defined in the process description.	URI	One (mandatory)
Data	The data provided for this input item.	Data structure, Table 23	Zero or one (conditional)[a]
Reference	HTTP resource that provides the input data.	Reference structure, Table 25	Zero or one (conditional)[a]

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Input	Nested input, child element.	DataInputType structure, Table 43 (this table)	Zero or more (conditional)[a]

[a] One and only one of these items shall be included.

Table 44 — Properties of the OutputDefinitionType

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
mimetype			
encoding	See Table 24 — Properties of the DataEncodingAttributes structure.		
schema			
transmission	Code that indicates the desired data transmission mode for this output.	Character string. Valid values are listed in the outputTransmission property of each Process Offering.	Zero or one (conditional)[a]
id	Identifier of a particular output, as defined in the process description.	URI	One (mandatory)
Output[b]	Nested output, child element.	OutputDefinitionType structure, Table 44 (this table).	Zero or more[c] (conditional)[d]

[a] This element may be omitted for (1) raw data output, (2) output elements that serve as nesting parents.

[b] Depending on the process model, a client may provide a basic specification of the desired output format. This information shall ensure a successful decoding of the process outputs. Use and interpretation of this element shall be specified within a WPS service profile.

[c] See Figure 2

[d] Include only for nested outputs.

10.9.2. Execute Response

Depending on the desired execution mode and the response type declared in the execute request, the execute response may take one of three different forms: A response document, a StatusInfo document, or raw data.

REQUIREMENTS CLASS 49	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model http://www.opengis.net/spec/WPS/2.0/req/service/model/status-info http://www.opengis.net/spec/WPS/2.0/req/service/model/result
NORMATIVE STATEMENTS	Requirement 49-1 *The execute response type shall comply with the properties specified in the execute request according to Table 45.* Requirement 49-2 *A raw data response shall comply with the output format specified in the execute request.* Requirement 49-3 _The response document shall comply with Requirements Class http://www.opengis.net/spec/WPS/2.0/req/service/model/result._ Requirement 49-4 _A StatusInfo response shall comply with Requirements Class http://www.opengis.net/spec/WPS/2.0/req/service/model/status-info._ Requirement 49-5 *The StatusInfo response to a valid execute response shall have the status element set to "Accepted".*

Table 45 — Possible responses to an execute request

		EXECUTION MODE		
		SYNC	ASYNC	AUTO
Response format	Raw	Raw data	StatusInfo document	Raw data or StatusInfo document (choice is made by server)[a]
	document	Response document	StatusInfo document	Response document or StatusInfo document (choice is made by server)[a]
	n/a	Response document (default)	StatusInfo document	

[a] The client identifies the server's choice by inspection of the response type (Table 45).

10.9.3. Execute Exceptions

When a WPS server encounters an error while performing an Execute operation, it shall return an exception report as specified in clause 8 of [OGC 06-121r9]. If appropriate, the server shall use additional exception codes as defined in this section.

For synchronous execution, an exception is returned instead of a result. For asynchronous execution, it is recommended to return the exception at the earliest possible time. In the case of a syntactically wrong request (e.g. due to the use of wrong identifiers and data formats), the exception report message may be returned instead of a StatusInfo document. If the exception

occurs later during execution, the StatusInfo shall be set to “Failed” and the GetResult operation shall be used to deliver the exception report.

REQUIREMENTS CLASS 50	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 50-1 *If a WPS server encounters an error while performing an Execute operation, it shall return an exception report message as specified in Clause 8 of [OGC 06-121r9].* Requirement 50-2 *If the encountered error is related to the properties and elements in the execute request or has occurred during the result computation, the server shall respond with the specific exception codes defined in Table 46.* Requirement 50-3 *If the server encounters an error for an asynchronously executed job, that had already been accepted, the server shall behave as follows: 1. The next StatusInfo responses shall have the status element set to “Failed”, and 2. The GetResult operation report shall return an exception report message as specified in Clause 8 of [OGC 06-121r9].*

Table 46 — Additional exception codes for the Execute operation

EXCEPTIONCODE VALUE	EXCEPTIONTEXT	LOCATOR	HTTP STATUS CODE
NoSuchProcess	One of the identifiers passed does not match with any of the processes offered by this server.	List of violating process identifiers.	400 (Bad request)
NoSuchMode	The process does not permit the desired execution mode.	Violating mode value.	400 (Bad request)
NoSuchInput	One or more of the input identifiers passed does not match with any of the input identifiers of this process.	List of violating input identifiers	400 (Bad request)
NoSuchOutput	One or more of the output identifiers passed does not match with any of the input identifiers of this process.	List of violating input identifiers	400 (Bad request)
DataNotAccessible	One of the referenced input data sets was inaccessible.	List of violating input identifiers	400 (Bad request)
SizeExceeded	The size of one of the input parameters was too large for this process to handle.	List of violating input identifiers	400 (Bad request)
TooManyInputs	Too many input items have been specified.	List of violating input identifiers	400 (Bad request)
TooManyOutputs	Too many output items have been specified.[a]	List of violating input identifiers	400 (Bad request)

EXCEPTIONCODE VALUE	EXCEPTIONTEXT	LOCATOR	HTTP STATUS CODE
ServerBusy	The server is too busy to accept and queue the request at this time.	None (omit locator parameter)	503 (Service Unavailable)
StorageNotSupported	Execute operation request included transmission= ”reference” for one of the outputs, but storage is not offered by this server.	None (omit locator parameter)	400 (Bad request)
NoSuchFormat	One or more of the input or output formats specified in the request did not match with any of the formats defined for that particular input or output.	List of violating input and / or output identifiers	400 (Bad request)
WrongInputData	One or more of inputs for which the service was able to retrieve the data but could not read it.	List of violating input identifiers	400 (Bad request)
InternalServerError	None (omit exception text)	None (omit locator parameter)	500 (Internal Server Error)[b]

[a] This shall be used in conjunction with raw data, where the execute request must specify only one output.

[b] If the cause could not be determined or is not covered by the above exception codes, this exception shall be used.

10.10. GetStatus Operation

The GetStatus operation allows WPS clients to query the status of an asynchronously executed job.

REQUIREMENTS CLASS 51	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 51-1 *Requirements class for the GetStatus request.* Requirement 51-2 *Requirements class for the GetStatus response.* Requirement 51-3 *Requirements class for GetStatus operation exceptions.*

10.10.1. GetStatus Request

The GetStatus request inherits basic properties from the RequestBaseType. It contains an additional element that identifies the JobID of the processing job, of which the status shall be returned.

REQUIREMENTS CLASS 52	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0 http://www.opengis.net/spec/WPS/2.0/req/service/model/handling
NORMATIVE STATEMENTS	Requirement 52-1 *GetStatus request shall comply with the structure defined in Figure 22 and Table 47.* Requirement 52-2 *The JobID used in the request shall be a valid identifier which the client has received with the execute response.*

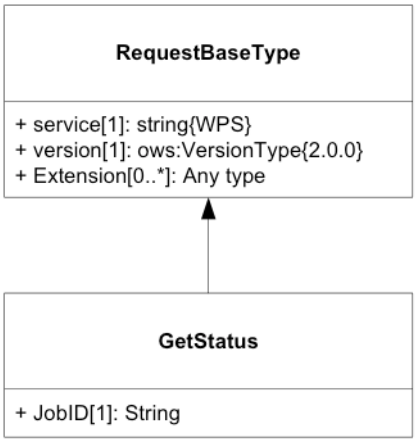

Figure 22 — GetStatus request UML class diagram

Table 47 — Additional properties in the GetStatus request

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
JobID	Job identifier.	Character String. This shall be a JobID the client has received during process execution.	One (mandatory)

10.10.2. GetStatus Response

The response to a GetStatus request is a StatusInfo document as defined in Clause 10.5.

REQUIREMENTS CLASS 53	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model http://www.opengis.net/spec/WPS/2.0/req/service/model/status-info
NORMATIVE STATEMENT	Requirement 53-1 _The response to a GetStatus request shall be a StatusInfo document as defined in http://www.opengis.net/spec/WPS/2.0/req/service/model/status-info._

10.10.3. GetStatus Exceptions

If a WPS server encounters an error while performing a GetStatus operation, it shall return an exception report as specified in Clause 8 of [OGC 06-121r9]. If the error was encountered due to an invalid process identifier, the server shall respond with the exception code defined in Table 48.

REQUIREMENTS CLASS 54	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 54-1 *If a WPS server encounters an error while performing a GetStatus operation, it shall return an exception report message as specified in Clause 8 of [OGC 06-121r9].* Requirement 54-2 *If the error was encountered due to an invalid JobID, the server shall respond with the exception code defined in Table 48.*

Table 48 — Additional exception codes for the GetStatus operation

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
NoSuch Job	The JobID from the request does not match any of the Jobs running on this server	Violating JobID	400 (Bad request)

10.11. GetResult Operation

The GetResult operation allows WPS clients to query the result of a finished processing job. It is used in conjunction with asynchronous execution.

REQUIREMENTS CLASS 55	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 55-1 *Requirements class for the GetResult request.* Requirement 55-2 *Requirements class for the GetResult response.* Requirement 55-3 *Requirements class for GetResult operation exceptions.*

10.11.1. GetResult Request

The GetResult request inherits basic properties from the RequestBaseType. It contains an additional element that identifies the JobID of the processing job, of which the result shall be returned.

REQUIREMENTS CLASS 56	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0 http://www.opengis.net/spec/WPS/2.0/req/service/model/handling
NORMATIVE STATEMENTS	Requirement 56-1 *The GetResult request shall comply with the structure defined in Figure 23 and Table 49.* Requirement 56-2 *The JobID used in the request shall be a valid identifier which the client has received with the execute response.*

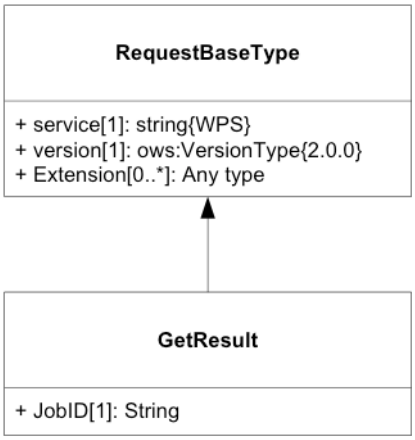

Figure 23 — GetResult request UML class diagram

Table 49 — Additional properties in the GetResult request

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
JobID	Job identifier.	Character String. This shall be a JobID the client has received during process execution.	One (mandatory)

10.11.2. GetResult Response

The response to a GetResult request is a Processing Result document as defined in Clause 10.6.

REQUIREMENTS CLASS 57	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model http://www.opengis.net/spec/WPS/2.0/req/service/model/processing-result
NORMATIVE STATEMENT	Requirement 57-1 _The response to a GetResult request shall be a Result document as defined in http://www.opengis.net/spec/WPS/2.0/req/service/model/processing-result._

10.11.3. GetResult Exceptions

When a WPS server encounters an error while performing a GetResult operation, it shall return an exception report as specified in Clause 8 of [OGC 06-121r9]. If the error was encountered due to an invalid JobID, the server shall respond with the exception code defined in Table 48

(NoSuchJob). If, for some reason, GetResult was invoked too early and the results have not been computed yet, the server shall respond with the exception defined in Table 50 (ResultNotReady).

If a job finishes with status "failed", GetResult will provide the corresponding exception report. In this case, the appropriate exception codes defined for the execute operation shall be used.

REQUIREMENTS CLASS 58	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 58-1 *If a WPS server encounters an error while performing a GetResult operation, it shall return an exception report message as specified in Clause 8 of [OGC 06-121r9].* Requirement 58-2 *If the error was encountered due to an invalid process identifier, the server shall respond with the exception code defined in Table 48 (NoSuchJob).* Requirement 58-3 *If the encountered error has occurred during the result computation, the server shall respond with the appropriate exception codes defined in Table 46.* Requirement 58-4 *If the results have not yet been computed and the job is still in the "running" state, the server shall respond with the exception code defined in Table 50.*

Table 50 — Additional exception codes for the GetResult operation

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
Result NotReady[a]	The result for the requested JobID has not yet been generated.	Violating JobID	400 (Bad request)

[a] Typically, this exception is thrown if a client violates the asynchronous protocol (Figure 4) and calls GetResult before the job execution has completed.

10.12. Synchronous WPS

The synchronous WPS is a requirements class profile that indicates the general availability of synchronous execution capabilities on a WPS server.

REQUIREMENTS CLASS 59	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/get-capabilities http://www.opengis.net/spec/WPS/2.0/req/service/model/describe-process http://www.opengis.net/spec/WPS/2.0/req/service/model/execute
NORMATIVE STATEMENTS	Requirement 59-1 *The Synchronous WPS server shall provide synchronous execution capabilities for one or more process offerings.* Requirement 59-2 *A Synchronous WPS server shall provide the following operations:* • *GetCapabilities* • *DescribeProcess* • *Execute (with mode "sync")*

10.13. Asynchronous WPS

The asynchronous WPS is a requirements class that indicates the general availability of asynchronous execution capabilities on a WPS server.

REQUIREMENTS CLASS 60	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/get-capabilities http://www.opengis.net/spec/WPS/2.0/req/service/model/describe-process http://www.opengis.net/spec/WPS/2.0/req/service/model/execute http://www.opengis.net/spec/WPS/2.0/req/service/model/get-status http://www.opengis.net/spec/WPS/2.0/req/service/model/get-result
NORMATIVE STATEMENTS	Requirement 60-1 *The Asynchronous WPS server shall provide asynchronous execution capabilities for one or more process offerings.* Requirement 60-2 *An Asynchronous WPS server shall provide the following operations:* • *GetCapabilities* • *DescribeProcess* • *Execute (with mode "async")* • *GetStatus* • *GetResult*

REQUIREMENTS CLASS 60

Requirement 60-3 *Result documents delivered by asynchronous execution shall contain a JobID.*

11

BINDING EXTENSIONS FOR WPS OPERATIONS

11 BINDING EXTENSIONS FOR WPS OPERATIONS

11.1. HTTP/POST + XML Binding

This section specifies the encoding of WPS requests using HTTP/POST and XML. The referred XML schema elements are provided by the associated schema package delivered with this standard and located at http://schemas.opengis.net/wps/2.0/.

REQUIREMENTS CLASS 61	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 61-1 *Requirements class for the GetCapabilities operation POST/XML encoding.* Requirement 61-2 *Requirements class for the DescribeProcess operation POST/XML encoding.* Requirement 61-3 *Requirements class for the Execute operation POST/XML encoding.* Requirement 61-4 *Requirements class for the GetStatus operation POST/XML encoding.* Requirement 61-5 *Requirements class for the GetResult operation POST/XML encoding.* Requirement 61-6 *Requirements class for the GetResult operation POST/XML encoding.* Requirement 61-7 *An operation extension for WPS servers that seeks compliance with this requirements class shall define an HTTP/POST+XML binding in compliance with [OGC 06-121r9].*

11.1.1. GetCapabilities

This clause specifies the XML encoding for the GetCapabilities operation.

A GetCapabilities example is listed in Annex B.4.

REQUIREMENTS CLASS 62	
OBLIGATION	requirement

REQUIREMENTS CLASS 62	
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/get-capabilities OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 62-1 *A GetCapabilities request using POST/XML shall be a valid XML document of the type wps:Capabilities.* Requirement 62-2 *The response to a successful GetCapabilities request shall be a valid XML document of the type wps:Capabilities.*

11.1.2. DescribeProcess

This clause specifies the XML encoding for the DescribeProcess operation.

A DescribeProcess example is listed in Annex B.5.

REQUIREMENTS CLASS 63	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/describe-process OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 63-1 *A DescribeProcess request using POST/XML shall be a valid XML document of the type wps:DescribeProcess.* Requirement 63-2 *The response to a successful DescribeProcess request shall be a valid XML document of the type wps:ProcessOfferings.*

11.1.3. Execute

This clause specifies the XML encoding for the Execute operation.

An Execute example is listed in Annex B.6.

REQUIREMENTS CLASS 64	
OBLIGATION	requirement
TARGET TYPE	Software implementation

REQUIREMENTS CLASS 64	
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/execute OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 64-1 *An Execute request using POST/XML shall be a valid XML document of the type wps:Execute.* Requirement 64-2 *The response to a successful Execute request shall is one of the following types:* 1. *A valid XML document of the type wps:Result* 2. *A valid XML document of the type wps:StatusInfo* 3. *A raw data response*

11.1.4. GetStatus

This clause specifies the XML encoding for the GetStatus operation.

A GetStatus example is listed in Annex B.7.

REQUIREMENTS CLASS 65	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/get-status OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 65-1 *A GetStatus request using POST/XML shall be a valid XML document of the type wps:GetStatus.* Requirement 65-2 *The response to a successful GetStatus request shall be a valid XML document of the type wps:StatusInfo.*

11.1.5. GetResult

This clause specifies the XML encoding for the GetResult operation.

A GetResult example is listed in Annex B.8.

REQUIREMENTS CLASS 66	
OBLIGATION	requirement
TARGET TYPE	Software implementation

REQUIREMENTS CLASS 66	
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/get-result OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 66-1 *A GetResult request using POST/XML shall be a valid XML document of the type wps:GetResult.* Requirement 66-2 *The response to a successful GetResult request shall be a valid XML document of the type wps:Result.*

11.2. HTTP/GET + KVP Binding

This section specifies the encoding of WPS requests using HTTP/GET and KVP. Since different HTTP client and server libraries may restrict the size of the URL this binding should only be used if the resulting HTTP/GET URL is reasonably short. The HTTP/GET + KVP binding is only defined for the operations GetCapabilities, DescribeProcess, GetStatus, and GetResult.[10].

Any WPS request in the HTTP/GET + KVP binding consists of a URL with KVP parameters. The response is always an XML document defined for the HTTP/POST + XML binding.

REQUIREMENTS CLASS 67	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model http://www.opengis.net/spec/WPS/2.0/req/service/binding/post-xml OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 67-1 *Requirements class for the GetCapabilities operation GET/KVP encoding.* Requirement 67-2 *Requirements class for the DescribeProcess operation GET/KVP encoding.* Requirement 67-3 *Requirements class for the GetStatus operation GET/KVP encoding.* Requirement 67-4 *Requirements class for the GetResult operation GET/KVP encoding.* Requirement 67-5 *Keys shall be case insensitive; values shall be case sensitive unless stated otherwise.* Requirement 67-6 *An operation extension for WPS servers that seeks compliance with this requirements class shall define an HTTP/GET+KVP binding in compliance with [OGC 06-121r9].*

[10]There is no HTTP/GET + KVP binding for execute requests due to the complex and nested structure of the request.

11.2.1. GetCapabilities

This clause specifies the KVP encoding for the GetCapabilities operation. A minimalist GetCapabilities request might look like this:

http://hostname:port/path?service=WPS&request=GetCapabilities

REQUIREMENTS CLASS 68	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/get-capabilities OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 68-1 *The KVP encoding of the GetCapabilities operation request shall be as specified in Table 5 in Subclause 7.2.3 of [OGC 06-121r9]. The fixed value of the service parameter shall be "WPS".* Requirement 68-2 *The response to a successful GetCapabilities request shall be a valid XML document of the type wps:Capabilities.*

11.2.2. DescribeProcess

This clause specifies the KVP encoding for the DescribeProcess operation. Possible DescribeProcess requests might look like these:

http://hostname:port/path?service=WPS&version=2.0.0&request=DescribeProcess&identifier=buffer,viewshed

http://hostname:port/path?service=WPS&version=2.0.0&request=DescribeProcess&identifier=ALL

REQUIREMENTS CLASS 69	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/describe-process OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 69-1 *The KVP encoding of DescribeProcess shall be as defined in Table 51.* Requirement 69-2 *The response to a successful GetCapabilities request shall be a valid XML document of the type wps:ProcessOfferings.*

Table 51 — DescribeProcess request KVP encoding

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
service	Identifier of the OGC service.	String, fixed to "WPS"	One (mandatory)
version	Request protocol version.	String, 2.0.0	One (mandatory)
request	Request name.	String, fixed to "DescribeProcess", case insensitive.	One (mandatory)
lang	Desired language of the process description.	IETF RFC 4646 language code of the human-readable text elements in the process description (e.g. "en"). This shall be one of the languages defined in the Capabilities document. If the client specifies more than one language, the server may create a response in any of these languages.	Zero or one (optional)
identifier	List of one or more process identifiers.	Comma-separated list of URI escaped character strings.[a] These shall be one or more of the process identifiers listed in the Process Summary elements in the Capabilities document. The fixed case insensitive value "ALL" indicates that the description of all processes shall be returned.	One (mandatory)

[a] The general syntax for URIs and URI escaping are defined in IETF RFC 3986.

11.2.3. GetStatus

This clause specifies the KVP encoding for the GetStatus operation. A possible GetStatus request might look like this:

http://hostname:port/path?service=WPS&version=2.0.0&request=GetStatus&jobid=FB6DD4B0-A2BB-11E3-A5E2-0800200C9A66

REQUIREMENTS CLASS 70	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/get-status OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 70-1 *The KVP encoding of GetStatus shall be as defined in Table 52.* Requirement 70-2 *The response to a successful GetStatus request shall be a valid XML document of the type wps:StatusInfo.*

Table 52 — GetStatus request KVP encoding

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
service	Identifier of the OGC service.	String, fixed to "WPS"	One (mandatory)
version	Request protocol version.	String, 2.0.0	One (mandatory)
request	Request name.	String, fixed to "GetStatus", case insensitive.	One (mandatory)
jobid	Job identifier.	Character String. This shall be a JobID the client has received during process execution.	One (mandatory)

11.2.4. GetResult

This clause specifies the KVP encoding for the GetResult operation. A possible GetResult request might look like this:

http://hostname:port/path?service=WPS&version=2.0.0&request=GetResult&jobid=FB6DD4B0-A2BB-11E3-A5E2-0800200C9A66

REQUIREMENTS CLASS 71	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/get-result OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 71-1 *The KVP encoding of GetResult shall be as defined in Table 53.* Requirement 71-2 *The response to a successful GetResult request shall be a valid XML document of the type wps:Result.*

Table 53 — GetResult request KVP encoding

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
service	Identifier of the OGC service.	String, fixed to "WPS"	One (mandatory)
version	Request protocol version.	String, 2.0.0	One (mandatory)

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
request	Request name.	String, fixed to “GetResult”, case insensitive.	One (mandatory)
jobid	Job identifier.	Character String. This shall be a JobID the client has received during process execution.	One (mandatory)

12

SERVICE PROFILES

12 SERVICE PROFILES

12.1. Basic WPS

The basic WPS is an aggregate requirements class that implements the WPS conceptual model based on the following requirements classes:

1. The WPS Service Model,
2. The binding extensions defined in Clause 10.12, and
3. The native WPS process model

REQUIREMENTS CLASS 72	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model http://www.opengis.net/spec/WPS/2.0/req/native-process/model http://www.opengis.net/spec/WPS/2.0/req/service/binding/post-xml http://www.opengis.net/spec/WPS/2.0/req/service/binding/get-kvp
NORMATIVE STATEMENTS	Requirement 72-1 _The Basic WPS server shall implement the common WPS service model as defined in Requirements class http://www.opengis.net/spec/WPS/2.0/req/service/model. _ Requirement 72-2 *The Basic WPS server shall support one or more of the following bindings for each operation:* 1. http://www.opengis.net/spec/WPS/2.0/req/service/binding/post-xml 2. http://www.opengis.net/spec/WPS/2.0/req/service/binding/get-kvp Requirement 72-3

13

DISMISS EXTENSION

13 DISMISS EXTENSION

The dismiss extension for WPS allows a client to communicate to the server that he is no longer interested in the results of a job. In this case, the server may free all associated resources and "forget" the JobID (Figure 24). For jobs that are still running, the server may cancel the execution at any time. For jobs that were already finished, the associated status information and the stored results may be deleted without further notice, regardless of the expiration time given in the last status report.

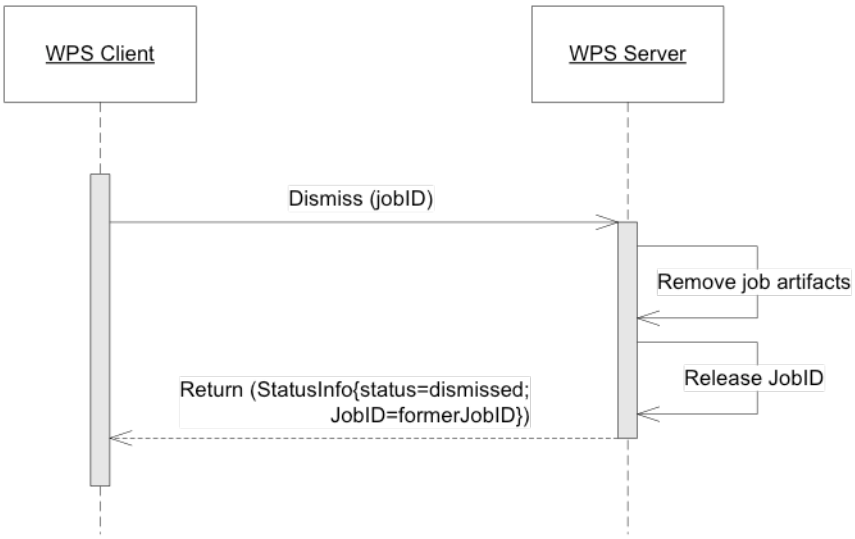

Figure 24 — Dismiss operation — UML sequence diagram

13.1. Dismiss Operation

The dismiss operation is a job control operation. Its availability is indicated per process using the jobControlOptions in the ProcessSummary and the ProcessOffering structures.

REQUIREMENTS CLASS 73	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0
NORMATIVE STATEMENTS	Requirement 73-1 *Requirements class for the Dismiss request.* Requirement 73-2 *Requirements class for the Dismiss response.* Requirement 73-3 *Requirements class for Dismiss operation exceptions.*

REQUIREMENTS CLASS 73	
	Requirement 73-4 *All process offerings that support the dismiss operation shall have the String "dismiss" listed in their jobControlOptions.* Requirement 73-5 *All results that can be dismissed after a process has finished shall have the JobID included.* Requirement 73-6 *The case insensitive status code for a dismissed job is "Dismissed".* Requirement 73-7 *The status "Dismissed" is transient. It shall only be used during a dismiss transaction.*

13.1.1. Dismiss Request

The Dismiss request inherits basic properties from the RequestBaseType. It contains an additional element that identifies the JobID of the processing job to be dismissed.

REQUIREMENTS CLASS 74	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0 http://www.opengis.net/spec/WPS/2.0/req/service/model/handling
NORMATIVE STATEMENTS	Requirement 74-1 *The Dismiss request shall comply with the structure defined in Figure 25 and Table 54.* Requirement 74-2 *The JobID used in the request shall be a valid identifier which the client has received with the execute response.*

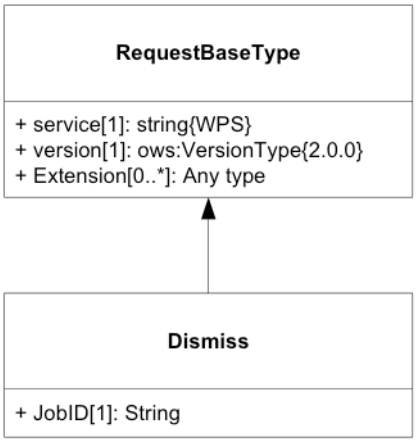

Figure 25 — Dismiss request UML class diagram

Table 54 — Additional properties in the Dismiss request

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
JobID	Job identifier.	Character String. This shall be a JobID the client has received during process execution.	One (mandatory)

13.1.2. Dismiss Response

The response to a Dismiss request is a StatusInfo document as defined in Clause 10.5. The status of the job shall be set to "Dismissed". Subsequent requests to the service using the dismissed jobID shall result in a NoSuchJob exception.

REQUIREMENTS CLASS 75	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITE	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model
NORMATIVE STATEMENTS	Requirement 75-1 _The response to a Dimiss request shall be a StatusInfo document as defined in http://www.opengis.net/spec/WPS/2.0/req/service/model/status-info._ Requirement 75-2 *The value of the Status element in the returned StatusInfo document shall be "dismissed".* Requirement 75-3 *With the delivery of a dismiss response, the associated JobID shall be released.*

13.1.3. Dismiss Exceptions

If a WPS server encounters an error while performing a Dismiss operation, it shall return an exception report as specified in Clause 8 of [OGC 06-121r9]. If the error was encountered due to an invalid JobID, the server shall respond with the exception code defined in Table 48 (NoSuchJob).

REQUIREMENTS CLASS 76	
OBLIGATION	requirement
TARGET TYPE	Derived encoding and software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/conceptual-model OWS Common 2.0

REQUIREMENTS CLASS 76	
NORMATIVE STATEMENTS	Requirement 76-1 *If a WPS server encounters an error while performing a GetStatus operation, it shall return an exception report message as specified in Clause 8 of [OGC 06-121r9].* Requirement 76-2 *If the error was encountered due to an invalid process identifier, the server shall respond with the exception code defined in Table 48 (NoSuchJob).*

13.2. Binding Extensions for the Dismiss Operation

13.2.1. HTTP/POST + XML Binding

This clause specifies the XML encoding for the Dismiss operation.

A Dismiss example is listed in Annex B.9.

REQUIREMENTS CLASS 77	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/dismiss http://www.opengis.net/spec/WPS/2.0/req/service/binding/post-xml
NORMATIVE STATEMENTS	Requirement 77-1 _The POST/XML encoding of Dismiss shall be in compliance with http://www.opengis.net/spec/WPS/2.0/req/service/binding/post-xml/extension-binding._ Requirement 77-2 *A Dismiss request using POST/XML shall be a valid XML document of the type wps:Dismiss.* Requirement 77-3 *The response to a successful Dismiss request shall be a valid XML document of the type wps:StatusInfo.*

13.2.2. HTTP/GET + KVP Binding

This clause specifies the KVP encoding for the Dismiss operation. A possible Dismiss request might look like this:

http://hostname:port/path?service=WPS&version=2.0.0&request=dismiss&jobid= FB6DD4B0-A2BB-11E3-A5E2-0800200C9A66

REQUIREMENTS CLASS 78	
OBLIGATION	requirement
TARGET TYPE	Software implementation
PREREQUISITES	http://www.opengis.net/spec/WPS/2.0/req/service/model/dismiss http://www.opengis.net/spec/WPS/2.0/req/service/binding/get-kvp
NORMATIVE STATEMENTS	Requirement 78-1 _The KVP encoding of Dismiss shall be in compliance with http://www.opengis.net/spec/WPS/2.0/req/service/binding/get-kvp/extension-binding._ Requirement 78-2 *The KVP encoding of Dismiss shall be as defined in Table 55.* Requirement 78-3 *The response to a successful Dismiss request shall be a valid XML document of the type wps:StatusInfo.*

Table 55 — Dismiss request KVP encoding

NAMES	DEFINITION	DATA TYPE AND VALUES	MULTIPLICITY AND USE
service	Identifier of the OGC service.	String, fixed to “WPS”	One (mandatory)
version	Request protocol version.	String, 2.0.0	One (mandatory)
request	Request name.	String, fixed to “Dismiss”, case insensitive.	One (mandatory)
jobid	Job identifier.	Character String. This shall be a JobID the client has received during process execution.	One (mandatory)

A

ANNEX A (NORMATIVE) ABSTRACT TEST SUITE

A ANNEX A (NORMATIVE) ABSTRACT TEST SUITE

Tests and requirement identifiers below are relative to **http://www.opengis.net/spec/WPS/2.0**

A.1. Basic WPS (Conformance Class)

The OGC URI identifier of this conformance class is: **http://www.opengis.net/spec/WPS/2.0/conf/service/profile/basic-wps**

Example

REQUIREMENT A.1	
STATEMENT	conf/service/profile/basic-wps

RECOMMENDATION A.1	
STATEMENT	Verify that the server implements the Basic WPS conformance class.

REQUIREMENT A.2	
STATEMENT	Verify that the server implements the Synchronous WPS and/or the Asynchronous WPS conformance class. Verify that the requests and responses to a supported operation are syntactically correct. Verify that the service supports the Synchronous WPS Conformance class, the Asynchronous WPS Conformance class or both. Verify that all process offerings implement the native process model. Verify the following list of conformance tests: • Annex A.4.1, Annex A.4.3 • Annex A.2 and/or Annex A.3

A.2. Synchronous WPS (Conformance Class)

The OGC URI identifier of this conformance class is: **http://www.opengis.net/spec/WPS/2.0/conf/service/synchronous-wps**

Example

REQUIREMENT A.3	
STATEMENT	conf/service/synchronous-wps

REQUIREMENT A.4	
STATEMENT	req/service/model/synchronous-wps/sync-execute

REQUIREMENT A.5	
STATEMENT	Verify that the server correctly advertises synchronous execution capabilities. Verify that the server correctly implements all operations that are mandatory for synchronous execution.

REQUIREMENT A.6	
STATEMENT	Verify that the GetCapabilities, DescribeProcess and Execute operations appear in the ows:OperationsMetadata element. Verify that the server offers at least one wps:ProcessSummary element whose “jobControlOptions” attribute contains “sync-execute”. Verify that the service supports at least one binding (e.g. HTTP GET/POST) for each supported operation. Verify the following list of conformance tests: • Annex A.5.1, Annex A.5.2, Annex A.5.3, Annex A.5.4, Annex A.5.5, Annex A.5.6, Annex A.5.7, Annex A.5.8 • Annex A.5.9 and/or Annex A.5.16 • Annex A.5.10 and/or Annex A.5.17 • Annex A.5.11, Annex A.5.13.

A.3. Asynchronous WPS (Conformance Class)

The OGC URI identifier of this conformance class is: **http://www.opengis.net/spec/WPS/2.0/conf/service/asynchronous-wps**

Example

REQUIREMENT A.7	
STATEMENT	conf/service/asynchronous-wps

REQUIREMENT A.8	
STATEMENT	req/service/model/asynchronous-wps

REQUIREMENT A.9	
STATEMENT	Verify that the server correctly advertises asynchronous execution capabilities. Verify that the server correctly implements all operations that are mandatory for asynchronous execution.

REQUIREMENT A.10	
STATEMENT	Verify that the GetCapabilities, DescribeProcess, Execute, GetStatus and GetResult operations appear in the ows:OperationsMetadata element. Verify that the server offers at least one wps:ProcessSummary element whose “jobControlOptions” attribute contains “async-execute”. Verify that the service supports at least one binding (e.g. HTTP GET/POST) for each supported operation. Verify the following list of conformance tests: • Annex A.5.1, Annex A.5.2, Annex A.5.3, Annex A.5.4, Annex A.5.5, Annex A.5.6, Annex A.5.7, Annex A.5.8 • Annex A.5.9 and/or Annex A.5.16 • Annex A.5.10 and/or Annex A.5.17 • Annex A.5.12, Annex A.5.13 • Annex A.5.14 and/or Annex A.5.18 • Annex A.5.15 and / or Annex A.5.19

A.4. WPS Process Model Encoding (Conformance Class)

The OGC URI identifier of this conformance class is:**http://www.opengis.net/spec/WPS/2.0/conf/process-model-encoding**

A.4.1. Process XML Encoding

Example

REQUIREMENT A.11	
STATEMENT	conf/process-model-encoding/xml-encoding/process

REQUIREMENT A.12	
STATEMENT	req/native-process/xml-encoding/process

REQUIREMENT A.13	
STATEMENT	Verify that a given process description is in compliance with the Process XML encoding.

REQUIREMENT A.14	
STATEMENT	Verify that the tested document fulfils all requirements listed in req/native-process/xml-encoding/process.

A.4.2. Generic Process XML Encoding

Example

REQUIREMENT A.15	
STATEMENT	conf/process-model-encoding/xml-encoding/generic-process

REQUIREMENT A.16	
STATEMENT	req/native-process/xml-encoding/generic-process

REQUIREMENT A.17	
STATEMENT	Verify that a given generic process description is in compliance with the generic process XML encoding.

REQUIREMENT A.18	
STATEMENT	Verify that the tested document fulfils all requirements listed in req/native-process/xml-encoding/generic-process.

A.4.3. Process data types XML Encoding

Example

REQUIREMENT A.19	
STATEMENT	conf/process-model-encoding/xml-encoding/datatypes

REQUIREMENT A.20	
STATEMENT	req/native-process/xml-encoding/datatypes

REQUIREMENT A.21	
STATEMENT	Verify that any XML data type description and values that are used in conjunction with the native process model are encoded in compliance with the process model XML encoding.

REQUIREMENT A.22	
STATEMENT	For **ComplexData** descriptions: Test passes if the tested XML fragment validates against *wps:ComplexData*. For LiteralData descriptions: Test passes if the tested XML fragment validates against wps:LiteralData. For **BoundingBoxData** descriptions: Test passes if the tested XML fragment validates against *wps:BoundingBoxData*. For **ComplexData values**: No general test available; the correctness of complex data values must be tested against a particular data type specification given by mime

REQUIREMENT A.22	
	type, encoding and schema. For **LiteralData values**: Test passes if the tested XML fragment validates against wps:LiteralValue. For **BoundingBoxData values**: Test passes if the tested XML fragment validates against *ows:BoundingBox*.

A.5. Basic tests

A.5.1. Request service name

Example

REQUIREMENT A.23	
STATEMENT	**conf/common-wps/handling/service**

REQUIREMENT A.24	
STATEMENT	**req/service/model/handling/service**

REQUIREMENT A.25	
STATEMENT	Verify that the correctly handles the service name parameter.

REQUIREMENT A.26	
STATEMENT	For each request type, send valid requests to server under test. Modulate service parameter: • Parameter value equal to what is required. Verify that request succeeds. • Parameter value not equal to what is required. Verify that request fails. Overall test passes if all individual tests pass.

A.5.2. Request version number

Example

REQUIREMENT A.27	
STATEMENT	conf/common-wps/handling/version

REQUIREMENT A.28	
STATEMENT	req/service/model/handling/version

REQUIREMENT A.29	
STATEMENT	Verify that the correctly handles the service version parameter.

REQUIREMENT A.30	
STATEMENT	For each request type, send valid requests to server under test. Modulate the version parameter: • Parameter value equal to what is required. Verify that request succeeds. • Parameter value not equal to what is required. Verify that request fails. Overall test passes if all individual tests pass.

A.5.3. Input data transmission by value

Example

REQUIREMENT A.31	
STATEMENT	conf/common-wps/data-transmission/input-by-value

REQUIREMENT A.32	
STATEMENT	req/conceptual-model/data-transmission/input-by-value

REQUIREMENT A.33	
STATEMENT	Verify that the server correctly handles input data transmission by value.

REQUIREMENT A.34	
STATEMENT	Send Execute requests to the server under test with valid inputs passed by value. Test passed if the execution finishes successfully.

A.5.4. Input data transmission by reference

Example

REQUIREMENT A.35	
STATEMENT	**conf/common-wps/input-by-reference**

REQUIREMENT A.36	
STATEMENT	**req/conceptual-model/data-transmission/input-by-reference**

REQUIREMENT A.37	
STATEMENT	Verify that the server correctly handles input data transmission by reference.

REQUIREMENT A.38	
STATEMENT	Send Execute requests to the server under test with valid inputs passed by reference. Test passed if the execution finishes successfully.

A.5.5. Output data transmission by value

Example

REQUIREMENT A.39	
STATEMENT	**conf/common-wps/data-transmission/output-by-value**

REQUIREMENT A.40	
STATEMENT	**req/conceptual-model/data-transmission/output-by-value**

REQUIREMENT A.41	
STATEMENT	Verify that the server correctly handles output data transmission by value.

REQUIREMENT A.42	
STATEMENT	Check the available process offerings for outputs that can be retrieved by value. If there is an output that can be retrieved by value, send an Execute request to the server requesting the output by value. Test passes if a valid Execute response is returned containing the requested output. Skip this test if no output can be retrieved by value.

A.5.6. Output data transmission by reference

Example

REQUIREMENT A.43	
STATEMENT	**conf/common-wps/data-transmission/output-by-reference**

REQUIREMENT A.44	
STATEMENT	**req/conceptual-model/data-transmission/output-by-reference**

REQUIREMENT A.45	
STATEMENT	Verify that the server correctly handles output data transmission by reference.

REQUIREMENT A.46	
STATEMENT	Check the available process offerings for outputs that can be retrieved by reference. If there is an output that can be retrieved by reference, send an Execute request to the server requesting the

REQUIREMENT A.46	
	output by reference. Test passes if a valid Execute response is returned containing a reference to the requested output. Skip this test if no output can be retrieved by reference.

A.5.7. Unique process identifier

Example

REQUIREMENT A.47	
STATEMENT	**conf/common-wps/identifier**

REQUIREMENT A.48	
STATEMENT	**req/conceptual-model/process/identifier**

REQUIREMENT A.49	
STATEMENT	Verify that each process the server offers has a unique identifier.

REQUIREMENT A.50	
STATEMENT	Get all available processes from the server under test. Test passes if all processes have a unique identifier.

A.5.8. Unique job identifier

Example

REQUIREMENT A.51	
STATEMENT	**conf/common-wps/job/identifier**

REQUIREMENT A.52	
STATEMENT	**req/conceptual-model/job/identifier**

REQUIREMENT A.53	
STATEMENT	Verify that the server creates a unique jobID for each job.

REQUIREMENT A.54	
STATEMENT	Send more than one asynchronous Execute requests to the server under test. Test passes if the retrieved JobIDs differ from each other.

A.5.9. GetCapabilities POST/XML encoding request/response

Example

REQUIREMENT A.55	
STATEMENT	**conf/service/binding/post-xml/get-capabilities/request-response**

REQUIREMENT A.56	
STATEMENT	**req/service/binding/post-xml/get-capabilities/request req/service/binding/post-xml/get-capabilities/response**

REQUIREMENT A.57	
STATEMENT	Verify that the server can handle GetCapabilities requests via POST/XML.

REQUIREMENT A.58	
STATEMENT	Send a valid GetCapabilities request to the server under test. Test passes if a valid document of the type *wps:Capabilities* is returned.

A.5.10. DescribeProcess POST/XML encoding request/response

Example

REQUIREMENT A.59	
STATEMENT	conf/service/binding/post-xml/describe-process/**request-response**

REQUIREMENT A.60	
STATEMENT	**req/service/binding/post-xml/describe-process/request req/service/binding/post-xml/describe-process/response**

REQUIREMENT A.61	
STATEMENT	Verify that the server can handle DescribeProcess requests via POST/XML.

REQUIREMENT A.62	
STATEMENT	Send a valid DescribeProcess request to the server under test. Test passes if a valid document of the type *wps:ProcessOfferings* is returned.

A.5.11. Synchronous Execute POST/XML encoding request/response

Example

REQUIREMENT A.63	
STATEMENT	conf/service/binding/post-xml/execute-sync/request-response

REQUIREMENT A.64	
STATEMENT	req/service/binding/post-xml/execute/request req/service/binding/post-xml/execute/response

REQUIREMENT A.65	
STATEMENT	Verify that the server can handle synchronous Execute requests via POST/XML.

REQUIREMENT A.66	
STATEMENT	Send a valid XML Execute request to the server under test, setting the “mode” attribute to “sync”. Modulate the “response” parameter: • Parameter value equal “document”. Verify that a valid Execute *wps:Result* is returned. • Parameter equal to “raw”. Verify that is returned. Overall test passes if all individual tests pass.

A.5.12. Asynchronous Execute POST/XML encoding request/response

Example

REQUIREMENT A.67	
STATEMENT	conf/service/binding/post-xml/execute-async/request-response

REQUIREMENT A.68	
STATEMENT	req/service/binding/post-xml/execute/request req/service/binding/post-xml/execute/response

REQUIREMENT A.69	
STATEMENT	Verify that the server can handle asynchronous Execute requests via POST/XML.

REQUIREMENT A.70	
STATEMENT	Send a valid XML Execute request to the server under test, setting the “mode” attribute to “async”. Test passes if a valid Execute *wps:StatusInfo* document is returned.

A.5.13. Auto Execute POST/XML encoding request/response

Example

REQUIREMENT A.71	
STATEMENT	conf/service/binding/post-xml/execute-auto/request-response

REQUIREMENT A.72	
STATEMENT	req/service/binding/post-xml/execute/request req/service/binding/post-xml/execute/response

REQUIREMENT A.73	
STATEMENT	Verify that the server can handle the execution mode “auto” requested via POST/XML.

REQUIREMENT A.74	
STATEMENT	Send a valid XML Execute request to the server under test, setting the “mode” attribute to “auto”. Modulate the “response” parameter. 1. If the process offering supports document output set “response” parameter value equal “document”. Check the execute response according to the following cases: a) If the process offering supports “sync-execute” and not “async-execute”: Verify that a valid Execute wps:Result document is returned. b) If the process offering supports “async-execute” and not “sync-execute”: Verify that a valid Execute wps:StatusInfo document is returned. c) If the process offering supports “sync-execute” and “async-execute”: Verify that a valid Execute wps:Result document or a valid wps:StatusInfo document is returned. 2. If the process offering supports raw output set “response” parameter equal to “raw”. Check the execute response according to the following cases: a) If the process offering supports “sync-execute” and not “async-execute”: Verify that valid that raw data is returned. b) If the process offering supports “async-execute” and not “sync-execute”: Verify that a valid Execute wps:StatusInfo document is returned. c) If the process offering supports “sync-execute” and “async-execute”: Verify that raw data or a valid wps:StatusInfo document is returned. Overall test passes if all individual tests pass.

A.5.14. GetStatus POST/XML encoding request/response

Example

REQUIREMENT A.75	
STATEMENT	conf/service/binding/post-xml/get-status/request-response

REQUIREMENT A.76	
STATEMENT	req/service/binding/post-xml/get-status/request req/service/binding/post-xml/get-status/response

REQUIREMENT A.77	
STATEMENT	Verify that the server can handle GetStatus requests via POST/XML.

REQUIREMENT A.78	
STATEMENT	Send a valid XML Execute request to the server under test, setting the “mode” attribute to “async”. Verify that a valid *wps:StatusInfo* document is returned. Extract the *wps:JobID*. Send a valid XML Get Status request to the server under test using the extracted JobID. Test passes if a valid *wps:StatusInfo* document is returned.

A.5.15. GetResult POST/XML encoding request/response

Example

REQUIREMENT A.79	
STATEMENT	conf/service/binding/get-kvp/describe-process/request-response

REQUIREMENT A.80	
STATEMENT	req/service/binding/get-kvp/describe-process/request req/service/binding/get-kvp/describe-process/response

REQUIREMENT A.81	
STATEMENT	Verify that the server can handle DescribeProcess requests via GET/KVP.

REQUIREMENT A.82	
STATEMENT	Send a valid KVP DescribeProcess request to the server under test, modulating upper and lower case of the parameter names. Test passes if a valid document of the type *wps:ProcessOfferings* is returned.

A.5.16. GetCapabilities GET/KVP encoding request/response

Example

REQUIREMENT A.83	
STATEMENT	conf/service/binding/get-kvp/get-capabilities/request-response

REQUIREMENT A.84	
STATEMENT	req/service/binding/get-kvp/get-capabilities/request req/service/binding/get-kvp/get-capabilities/response

REQUIREMENT A.85	
STATEMENT	Verify that the server can handle GetCapabilities requests via GET/KVP.

REQUIREMENT A.86	
STATEMENT	Send a valid KVP GetCapabilities request to the server under test, modulating upper and lower case of the parameter names. Test passes if a valid document of the type *wps:Capabilities* is returned.

A.5.17. DescribeProcess GET/KVP encoding request/response

Example

REQUIREMENT A.87	
STATEMENT	conf/service/binding/get-kvp/describe-process/request-response

REQUIREMENT A.88	
STATEMENT	req/service/binding/get-kvp/describe-process/request req/service/binding/get-kvp/describe-process/response

REQUIREMENT A.89	
STATEMENT	Verify that the server can handle DescribeProcess requests via GET/KVP.

REQUIREMENT A.90	
STATEMENT	Send a valid KVP DescribeProcess request to the server under test, modulating upper and lower case of the parameter names. Test passes if a valid document of the type *wps:ProcessOfferings* is returned.

A.5.18. GetStatus GET/KVP encoding request/response

Example

REQUIREMENT A.91	
STATEMENT	**conf/service/binding/get-kvp/get-status/request-response**

REQUIREMENT A.92	
STATEMENT	**req/service/binding/get-kvp/get-status/request req/service/binding/get-kvp/get-status/response**

REQUIREMENT A.93	
STATEMENT	Verify that the server can handle GetStatus requests via GET/KVP.

REQUIREMENT A.94	
STATEMENT	Send a valid XML Execute request to the server under test, setting the “mode” attribute to “async”. Verify that a valid *wps:StatusInfo* document is returned. Extract the *wps:JobID*. Send a valid KVP Get Status request to the server under test, using the extracted JobID and modulating upper and lower case of the parameter names. Test passes if a valid document of the type *wps:StatusInfo* is returned.

A.5.19. GetResult GET/KVP encoding request/response

Example

REQUIREMENT A.95	
STATEMENT	conf/service/binding/get-kvp/get-result/request-response

REQUIREMENT A.96	
STATEMENT	req/service/binding/get-kvp/get-result/request req/service/binding/get-kvp/get-result/response

REQUIREMENT A.97	
STATEMENT	Verify that the server can handle GetResult requests via GET/KVP.

REQUIREMENT A.98	
STATEMENT	Send a valid XML Execute request to the server under test, setting the "mode" attribute to "async". Modulate the "response" parameter. Verify that a valid *wps:StatusInfo* document is returned. Extract the *wps:JobID*. Check the status of the job. If the job succeeded, send a valid KVP GetResult request to the server under test using the extracted JobID and modulating upper and lower case of the parameter names. Depending on the value of the "response" parameter of the above Execute request: • Parameter value equal "document". Verify that a valid Execute *wps:Result* document is returned. • Parameter equal to "raw". Verify that raw is returned. Overall test passes if all individual tests pass.

A.6. Dismiss Extension (Conformance Class)

The OGC URI identifier of this conformance class is: **http://www.opengis.net/spec/WPS/2.0/conf/service/dismiss-extension**

A.6.1. Dismiss POST/XML encoding request/response

Example

REQUIREMENT A.99	
STATEMENT	conf/service/binding/post-xml/dismiss/request-response

REQUIREMENT A.100	
STATEMENT	req/service/binding/post-xml/dismiss/request req/service/binding/post-xml/dismiss/response

REQUIREMENT A.101	
STATEMENT	Verify that the server can handle Dismiss requests via POST/XML.

REQUIREMENT A.102	
STATEMENT	Precondition: The process offering used for testing must have “dismiss” listed among its job control options. Send a valid XML Execute request to the server under test, setting the “mode” attribute to “async”. Verify that a valid *wps:StatusInfo* document is returned. Extract the *wps:JobID*. Send a valid XML Dismiss request to the server under test using the extracted JobID. Test passes if a valid *wps:StatusInfo* document is returned containing a *wps:Status* element with value “Dismissed” (case insensitive).

A.6.2. Dismiss GET/KVP encoding request/response

Example

REQUIREMENT A.103	
STATEMENT	conf/service/binding/get-kvp/dismiss/request-response

REQUIREMENT A.104	
STATEMENT	req/service/binding/get-kvp/dismiss/request req/service/binding/get-kvp/dismiss/response

REQUIREMENT A.105

STATEMENT	Verify that the server can handle Dismiss requests via GET/KVP.

REQUIREMENT A.106

STATEMENT	Precondition: The process offering used for testing must have “dismiss” listed among its job control options. Send a valid XML Execute request to the server under test, setting the “mode” attribute to “async”. Verify that a valid *wps:StatusInfo* document is returned. Extract the *wps:JobID*. Send a valid KVP Dismiss request to the server under test using the extracted JobID and modulating upper and lower case of the parameter names. Test passes if a valid document of the type *wps:StatusInfo* document is returned containing a *wps:Status* element with value “Dismissed” (case insensitive).

B

ANNEX B (INFORMATIVE) XML EXAMPLES

B ANNEX B (INFORMATIVE) XML EXAMPLES

B.1. Data Types

B.1.1. Complex Data Description

```
<wps:ComplexData>
  <wps:Format mimeType="application/geotiff" encoding="raw"
default="true"/>
  <wps:Format mimeType="application/geotiff" encoding="base64"/>
</wps:ComplexData>
```

Listing

B.1.2. Literal Data Description

```
<wps:LiteralData>
  <wps:Format mimeType="text/plain" default="true"/>
  <wps:Format mimeType="text/xml"/>
  <LiteralDataDomain default="true">
    <ows:AllowedValues>
      <ows:Range>
        <ows:MinimumValue>1</ows:MinimumValue>
        <ows:MaximumValue>1000</ows:MaximumValue>
      </ows:Range>
    </ows:AllowedValues>
    <ows:DataType
ows:reference="http://www.w3.org/2001/XMLSchema#float">float
</ows:DataType>
    <ows:UOM>meters</ows:UOM>
    <ows:DefaultValue>100</ows:DefaultValue>
  </LiteralDataDomain>
  <LiteralDataDomain>
    <ows:AllowedValues>
      <ows:Range>
        <ows:MinimumValue>1</ows:MinimumValue>
        <ows:MaximumValue>3000</ows:MaximumValue>
      </ows:Range>
    </ows:AllowedValues>
    <ows:DataType
ows:reference="http://www.w3.org/2001/XMLSchema#float">float
</ows:DataType>
```

```
    <ows:UOM>feet</ows:UOM>
  </LiteralDataDomain>
</wps:LiteralData>
```

Listing

B.1.3. Literal data values

```
<LiteralValue
  dataType=http://www.w3.org/2001/XMLSchema#double
  uom="meter">
    42.1
</LiteralValue>

<LiteralValue
  dataType="http://www.w3.org/2001/XMLSchema#string">
    ArableLand
</LiteralValue>
```

Listing

B.1.4. BoundingBox Data Description

```
<wps:BoundingBoxData>
  <wps:Format mimeType="text/plain" default="true"/>
  <wps:Format mimeType="text/xml"/>
  <wps:SupportedCRS default="true">EPSG:4326</wps:SupportedCRS>
  <wps:SupportedCRS>
    http://www.opengis.net/def/crs/EPSG/0/4258
  </wps:SupportedCRS>
</wps:BoundingBoxData>
```

Listing

B.1.5. BoundingBox Data Values

```
<ows:BoundingBox crs="EPSG:4326">
  <ows:LowerCorner>51.9 7.0</ows:LowerCorner>
  <ows:UpperCorner>53.0 8.0</ows:UpperCorner>
</ows:BoundingBox>

<ows:BoundingBox
   crs="http://www.opengis.net/def/crs/EPSG/0/4258">
  <ows:LowerCorner>51.9 7.0</ows:LowerCorner>
  <ows:UpperCorner>53.0 8.0</ows:UpperCorner>
</ows:BoundingBox>
```

Listing

B.2. Process Description

This example describes a buffer command that accepts polygon coordinates in GML, and used a buffer distance in meters to produce a buffered polygon feature, which is output in GML.

```
<wps:Process
  xmlns:wps="http://www.opengis.net/wps/2.0"
  xmlns:ows="http://www.opengis.net/ows/2.0"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation=
"http://www.opengis.net/wps/2.0 ../../wps.xsd">

  <ows:Title>Planar Buffer operation for GML features</ows:Title>
  <ows:Abstract>
  Create a buffer around a GML feature. Accepts any valid GML
  feature and computes the joint buffer.</ows:Abstract>
  <ows:Identifier>
    http://some.host/profileregistry/implementation/Planar-GML-
Buffer
  </ows:Identifier>
  <ows:Metadata
    xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process-profile/concept"
    xlink:href="http://some.host/profileregistry/concept/
buffer"/>
  <ows:Metadata
    xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process-profile/concept"
    xlink:href="http://some.host/profileregistry/concept/
planarbuffer"/>
  <ows:Metadata
    xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process-profile/generic"
    xlink:href="http://some.host/profileregistry/generic/
SF-Buffer"/>
  <ows:Metadata
    xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process/description/documentation"
    xlink:href="http://some.host/profileregistry/implementation/
Planar-GML-Buffer.html"/>
  <wps:Input>
    <ows:Title>Geometry to be buffered</ows:Title>
    <ows:Abstract>Geometry input in GML</ows:Abstract>
    <ows:Identifier>INPUT_GEOMETRY</ows:Identifier>
    <ows:Metadata
      xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process/description/documentation"
      xlink:href="http://some.host/profileregistry/
implementation/Planar-GML-Buffer.html#input_geometry"/>
    <wps:ComplexData>
      <wps:Format mimeType="text/xml" encoding="UTF-8"
        schema="http://schemas.opengis.net/gml/3.2.1/feature.xsd"
        default="true"/>
    </wps:ComplexData>
  </wps:Input>
  <wps:Input>
    <ows:Title>Distance</ows:Title>
    <ows:Abstract>
```

```
      Distance to be used to calculate buffer.
    </ows:Abstract>
    <ows:Identifier>DISTANCE</ows:Identifier>
    <ows:Metadata
      xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process/description/documentation"
      xlink:href="http://some.host/profileregistry/
implementation/Planar-GML-Buffer.html#distance"/>
    <wps:LiteralData>
      <wps:Format mimeType="text/plain" default="true"/>
      <wps:Format mimeType="text/xml"/>
      <LiteralDataDomain default="true">
        <ows:AllowedValues>
          <ows:Range>
            <ows:MinimumValue>-INF</ows:MinimumValue>
            <ows:MaximumValue>INF</ows:MaximumValue>
          </ows:Range>
        </ows:AllowedValues>
        <ows:DataType
          ows:reference="http://www.w3.org/2001/
XMLSchema#double">Double</ows:DataType>
      </LiteralDataDomain>
    </wps:LiteralData>
  </wps:Input>
  <wps:Output>
    <ows:Title>Buffered Geometry</ows:Title>
    <ows:Abstract>
      GML stream describing the buffered Geometry.</ows:Abstract>
    <ows:Identifier>BUFFERED_GEOMETRY</ows:Identifier>
    <ows:Metadata
      xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process/description/documentation"
      xlink:href="http://some.host/profileregistry/
implementation/Planar-GML-Buffer.html#buffered_geometry"/>
    <wps:ComplexData>
      <wps:Format mimeType="text/xml" encoding="UTF-8"
        schema="http://schemas.opengis.net/gml/3.2.1/feature.xsd"
        default="true"/>
    </wps:ComplexData>
  </wps:Output>
</wps:Process>
```

Listing

B.3. Generic Process

This example describes a generic profile for a simple features buffer. It returns a geometry that represents all points whose distance from this Geometry is less than or equal to distance. Calculations are in the Spatial Reference System of this Geometry.

```
<wps:GenericProcess
  xmlns:ows="http://www.opengis.net/ows/2.0"
  xmlns:wps="http://www.opengis.net/wps/2.0"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  xmlns:xml="http://www.w3.org/XML/1998/namespace"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../../wps.xsd">
  <ows:Title>Simple Features Buffer</ows:Title>
```

```
  <ows:Identifier>http://some.host/profileregistry/generic/
    SF-Buffer</ows:Identifier>
  <ows:Metadata
    xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process-profile/concept"
    xlink:href="http://some.host/profileregistry/
concept/buffer"/>
  <ows:Metadata
    xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process-profile/concept"
    xlink:href="http://some.host/profileregistry/
concept/planarbuffer"/>

  <ows:Metadata
    xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process/description/documentation"
    xlink:href="http://some.host/profileregistry/
generic/SF-Buffer.html"/>
  <wps:Input>
    <ows:Title>Input Geometry</ows:Title>
    <ows:Identifier>INPUT_GEOMETRY</ows:Identifier>
    <ows:Metadata
      xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process/description/documentation"
      xlink:href="http://some.host/profileregistry/
generic/SF-Buffer.html#input_geometry"/>
  </wps:Input>
  <wps:Input>
    <ows:Title>Distance</ows:Title>
    <ows:Identifier>DISTANCE</ows:Identifier>
    <ows:Metadata
      xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process/description/documentation"
      xlink:href="http://some.host/profileregistry/
generic/SF-Buffer.html#distance"/>
  </wps:Input>
  <wps:Output>
    <ows:Title>Buffered Geometry</ows:Title>
    <ows:Identifier>BUFFERED_GEOMETRY</ows:Identifier>
    <ows:Metadata
      xlink:role="http://www.opengis.net/spec/wps/2.0/def/
process/description/documentation"
      xlink:href="http://some.host/profileregistry/
generic/SF-Buffer.html#buffered_geometry"/>
  </wps:Output>
</wps:GenericProcess>
```

Listing

B.4. GetCapabilities

B.4.1. GetCapabilities Request

```
<wps:GetCapabilities service="WPS"
  xmlns:ows="http://www.opengis.net/ows/2.0"
  xmlns:wps="http://www.opengis.net/wps/2.0"
```

```
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../wps.xsd">
</wps:GetCapabilities>
```

Listing

B.4.2. GetCapabilities Response

```
<wps:Capabilities service="WPS" version="2.0.0"
  xmlns:ows="http://www.opengis.net/ows/2.0"
  xmlns:wps="http://www.opengis.net/wps/2.0"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  xmlns:xml="http://www.w3.org/XML/1998/namespace"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../wps.xsd ">
  <ows:ServiceIdentification>
    <ows:Title>MyWebProcessingService</ows:Title>
    <ows:Abstract>
      A Web Processing Service offering typical GIS distance
      transform processes.
    </ows:Abstract>
    <ows:Keywords>
      <ows:Keyword>Geoprocessing</ows:Keyword>
      <ows:Keyword>Toolbox</ows:Keyword>
      <ows:Keyword>Distance transform</ows:Keyword>
    </ows:Keywords>
    <ows:ServiceType>WPS</ows:ServiceType>
    <ows:ServiceTypeVersion>2.0.0</ows:ServiceTypeVersion>
    <ows:Fees>NONE</ows:Fees>
    <ows:AccessConstraints>NONE</ows:AccessConstraints>
  </ows:ServiceIdentification>
  <ows:ServiceProvider>
    <ows:ProviderName>TU Dresden</ows:ProviderName>
    <ows:ProviderSite
xlink:href="http://tu-dresden.de/geo/gis" />
    <ows:ServiceContact>
      <ows:IndividualName>Matthias Mueller</ows:IndividualName>
      <ows:ContactInfo>
        <ows:Address>
          <ows:ElectronicMailAddress>
            matthias_mueller@tu-dresden.de
          </ows:ElectronicMailAddress>
        </ows:Address>
      </ows:ContactInfo>
    </ows:ServiceContact>
  </ows:ServiceProvider>
  <ows:OperationsMetadata>
    <ows:Operation name="GetCapabilities">
      <ows:DCP>
        <ows:HTTP>
          <ows:Get
          xlink:href="http://wps1.gis.geo.tu-dresden.de/wps"/>
        </ows:HTTP>
      </ows:DCP>
    </ows:Operation>
    <ows:Operation name="DescribeProcess">
      <ows:DCP>
        <ows:HTTP>
          <ows:Get
          xlink:href="http://wps1.gis.geo.tu-dresden.de/wps"/>
```

```
          <ows:Post
          xlink:href="http://wps1.gis.geo.tu-dresden.de/wps"/>
        </ows:HTTP>
      </ows:DCP>
    </ows:Operation>
    <ows:Operation name="Execute">
      <ows:DCP>
        <ows:HTTP>
          <ows:Post
          xlink:href="http://wps1.gis.geo.tu-dresden.de/wps"/>
        </ows:HTTP>
      </ows:DCP>
    </ows:Operation>
    <ows:Operation name="GetStatus">
      <ows:DCP>
        <ows:HTTP>
          <ows:Get
          xlink:href="http://wps1.gis.geo.tu-dresden.de/wps"/>
          <ows:Post
          xlink:href="http://wps1.gis.geo.tu-dresden.de/wps"/>
        </ows:HTTP>
      </ows:DCP>
    </ows:Operation>
    <ows:Operation name="GetResult">
      <ows:DCP>
        <ows:HTTP>
          <ows:Get
          xlink:href="http://wps1.gis.geo.tu-dresden.de/wps"/>
          <ows:Post
          xlink:href="http://wps1.gis.geo.tu-dresden.de/wps"/>
        </ows:HTTP>
      </ows:DCP>
    </ows:Operation>
    <ows:Operation name="Dismiss">
      <ows:DCP>
        <ows:HTTP>
          <ows:Get
          xlink:href="http://wps1.gis.geo.tu-dresden.de/wps"/>
          <ows:Post
          xlink:href="http://wps1.gis.geo.tu-dresden.de/wps"/>
        </ows:HTTP>
      </ows:DCP>
    </ows:Operation>
  </ows:OperationsMetadata>
  <wps:Contents>
    <wps:ProcessSummary
jobControlOptions="sync-execute async-execute dismiss">
      <ows:Title>Euclidean Distance</ows:Title>
      <ows:Identifier>
        http://my.site/distance-transform/euclidean-distance
      </ows:Identifier>
    </wps:ProcessSummary>
    <wps:ProcessSummary
jobControlOptions="async-execute dismiss">
      processVersion="1.4.0">
      <ows:Title>Cost Distance</ows:Title>
      <ows:Identifier>
        http://my.site/distance-transform/cost-distance
      </ows:Identifier>
    </wps:ProcessSummary>
  </wps:Contents>
```

```
</wps:Capabilities>
```

Listing B.1

B.5. DescribeProcess

B.5.1. DescribeProcess Request

```
<wps:DescribeProcess service="WPS" version="2.0.0"
  xmlns:ows="http://www.opengis.net/ows/2.0"
  xmlns:wps="http://www.opengis.net/wps/2.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../wps.xsd">
  <ows:Identifier>Buffer</ows:Identifier>
  <ows:Identifier>Viewshed</ows:Identifier>
</wps:DescribeProcess>
```

Listing

B.5.2. DescribeProcess Response

```
<wps:ProcessOfferings xmlns:wps="http://www.opengis.net/wps/2.0.0"
  xmlns:ows=http://www.opengis.net/ows/2.0
  xmlns:xlink="http://www.w3.org/1999/xlink"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../wps.xsd">
  <wps:ProcessOffering
jobControlOptions="sync-execute async-execute dismiss"
outputTransmission="value reference">
    <wps:Process>
      <ows:Title>
        Planar Buffer operation for Simple Features
      </ows:Title>
      <ows:Abstract>
        Create a buffer around Simple Feature geometries. Accepts
        any valid simple features input in GML or GeoJson and
        computes their joint buffer geometry.
      </ows:Abstract>
      <ows:Identifier>
        http://my.wps.server/processes/proximity/Planar-Buffer
      </ows:Identifier>
      <ows:Metadata
        xlink:role="http://www.opengis.net/spec/wps/2.0/
def/process-profile/concept"
        xlink:href="http://some.host/profileregistry/
concept/buffer"/>
      <ows:Metadata
        xlink:role="http://www.opengis.net/spec/wps/2.0/
def/process-profile/concept"
        xlink:href="http://some.host/profileregistry/
concept/planarbuffer"/>
      <ows:Metadata
        xlink:role="http://www.opengis.net/spec/wps/2.0/
def/process-profile/generic"
```

```
        xlink:href="http://some.host/profileregistry/
generic/SF-Buffer"/>
      <ows:Metadata
        xlink:role="http://www.opengis.net/spec/wps/2.0/
def/process/description/documentation"
        xlink:href="http://my.wps.server/processes/
proximity/Planar-Buffer.html"/>
      <wps:Input>
        <ows:Title>Geometry to be buffered</ows:Title>
        <ows:Abstract>
          Simple Features geometry input in GML or GeoJson
        </ows:Abstract>
        <ows:Identifier>INPUT_GEOMETRY</ows:Identifier>
        <ows:Metadata
          xlink:role="http://www.opengis.net/spec/wps/2.0/
def/process/description/documentation"
          xlink:href="http://my.wps.server/processes/
proximity/Planar-Buffer.html#input_geometry"/>
        <wps:ComplexData>
          <wps:Format mimeType="text/xml" encoding="UTF-8"
            schema="http://schemas.opengis.net/gml/
3.2.1/feature.xsd" default="true"/>
          <wps:Format mimeType="application/json"
encoding="UTF-8"/>
        </wps:ComplexData>
      </wps:Input>
      <wps:Input>
        <ows:Title>Distance</ows:Title>
        <ows:Abstract>
           Distance to be used to calculate buffer.
        </ows:Abstract>
        <ows:Identifier>DISTANCE</ows:Identifier>
        <ows:Metadata
          xlink:role="http://www.opengis.net/spec/wps/2.0/
def/process/description/documentation"
          xlink:href="http://my.wps.server/processes/
proximity/Planar-Buffer.html#distance"/>
        <wps:LiteralData>
          <wps:Format mimeType="text/plain" default="true"/>
          <wps:Format mimeType="text/xml"/>
          <LiteralDataDomain default="true">
            <ows:AllowedValues>
              <ows:Range>
                <ows:MinimumValue>-INF</ows:MinimumValue>
                <ows:MaximumValue>INF</ows:MaximumValue>
              </ows:Range>
            </ows:AllowedValues>
            <ows:DataType
              ows:reference="http://www.w3.org/2001/
XMLSchema#double">
              Double
            </ows:DataType>
          </LiteralDataDomain>
        </wps:LiteralData>
      </wps:Input>
      <wps:Output>
        <ows:Title>Buffered Geometry</ows:Title>
        <ows:Abstract>
          Output Geometry in GML or GeoJson
        </ows:Abstract>
        <ows:Identifier>BUFFERED_GEOMETRY</ows:Identifier>
        <ows:Metadata
          xlink:role="http://www.opengis.net/spec/wps/2.0/
```

```
def/process/description/documentation"
          xlink:href="http://my.wps.server/processes/
proximity/Planar-Buffer.html#buffered_geometry"/>
        <wps:ComplexData>
          <wps:Format mimeType="text/xml" encoding="UTF-8"
            schema="http://schemas.opengis.net/gml/
3.2.1/feature.xsd" default="true"/>
          <wps:Format mimeType="application/json"
encoding="UTF-8"/>
        </wps:ComplexData>
      </wps:Output>
    </wps:Process>
  </wps:ProcessOffering>
</wps:ProcessOfferings>
```

Listing B.2

B.6. Execute

B.6.1. Execute Request (asynchronous, result document)

```
<wps:Execute
  xmlns:wps="http://www.opengis.net/wps/2.0"
  xmlns:ows="http://www.opengis.net/ows/2.0"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../wps.xsd"
  service="WPS" version="2.0.0" response="document" mode="async">
  <ows:Identifier>
    http://my.wps.server/processes/proximity/Planar-Buffer
  </ows:Identifier>
  <wps:Input id="INPUT_GEOMETRY">
    <wps:Reference xlink:href="http://some.data.server/
mygmldata.xml"/>
  </wps:Input>
  <wps:Input id="DISTANCE">
    <wps:Data>10</wps:Data>
  </wps:Input>
  <!- Uses default output format ->
  <wps:Output id="BUFFERED_GEOMETRY"
wps:dataTransmissionMode="reference">
  </wps:Output>
</wps:Execute>
```

Listing

B.6.2. Execute Response (StatusInfo)

```
<wps:StatusInfo xmlns:ows="http://www.opengis.net/ows/2.0"
  xmlns:wps="http://www.opengis.net/wps/2.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../wps.xsd">
  <wps:JobID>FB6DD4B0-A2BB-11E3-A5E2-0800200C9A66</wps:JobID>
  <wps:Status>Accepted</wps:Status>
```

```
  <wps:NextPoll>2014-12-24T16:00:00Z</wps:NextPoll>
</wps:StatusInfo>
```

Listing

B.6.3. Execute Response (Result)

```
<wps:Result
  xmlns:ows="http://www.opengis.net/ows/2.0"
  xmlns:wps="http://www.opengis.net/wps/2.0"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../wps.xsd ">
  <wps:JobID>FB6DD4B0-A2BB-11E3-A5E2-0800200C9A66</wps:JobID>
  <wps:ExpirationDate>2014-12-24T24:00:00Z</wps:ExpirationDate>
  <wps:Output id="BUFFERED_GEOMETRY">
    <wps:Reference
      xlink:href="http://result.data.server/
FB6DD4B0-A2BB-11E3-A5E2-0800200C9A66/
BUFFERED_GEOMETRY.xml"/>
  </wps:Output>
</wps:Result>
```

Listing

B.7. GetStatus

B.7.1. GetStatus Request

```
<wps:GetStatus service="WPS" version="2.0.0"
  xmlns:wps="http://www.opengis.net/wps/2.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../wps.xsd ">
  <wps:JobID>FB6DD4B0-A2BB-11E3-A5E2-0800200C9A66</wps:JobID>
</wps:GetStatus>
```

Listing

B.7.2. GetStatus Response

```
<wps:StatusInfo xmlns:ows="http://www.opengis.net/ows/2.0"
  xmlns:wps="http://www.opengis.net/wps/2.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../wps.xsd">
  <wps:JobID>FB6DD4B0-A2BB-11E3-A5E2-0800200C9A66</wps:JobID>
  <wps:Status>Running</wps:Status>
  <wps:NextPoll>2014-12-24T16:00:00Z</wps:NextPoll>
</wps:StatusInfo>
```

Listing

B.8. GetResult

B.8.1. GetResult Request

```
<wps:GetResult service="WPS" version="2.0.0"
  xmlns:wps="http://www.opengis.net/wps/2.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../wps.xsd ">
  <wps:JobID>FB6DD4B0-A2BB-11E3-A5E2-0800200C9A66</wps:JobID>
</wps:GetResult>
```

Listing

B.8.2. GetResult Response

See Annex B.6.3.

B.9. Dismiss

B.9.1. Dismiss Request

```
<wps:Dismiss service="WPS" version="2.0.0"
  xmlns:wps="http://www.opengis.net/wps/2.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../wps.xsd ">
  <wps:JobID>FB6DD4B0-A2BB-11E3-A5E2-0800200C9A66</wps:JobID>
</wps:Dismiss>
```

Listing

B.9.2. Dismiss Response

```
<wps:StatusInfo xmlns:ows="http://www.opengis.net/ows/2.0"
  xmlns:wps="http://www.opengis.net/wps/2.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.opengis.net/wps/2.0 ../wps.xsd">
  <wps:JobID>FB6DD4B0-A2BB-11E3-A5E2-0800200C9A66</wps:JobID>
  <wps:Status>Dismissed</wps:Status>
</wps:StatusInfo>
```

Listing

B.10. Profile inheritance example

Taking the example of a simple Mosaic process, Figure B.1 and Figure B.2 illustrate the inheritance rules for process profiles provided in Clause 8.5.4. In its most simple form, a mosaic process transforms a set if tiles (Coverages or georeferenced imagery) into a single output dataset. Starting from the conceptual level, the process definition is incrementally refined until the specificity of an implementation profile is reached. At the level of a profile implementation, e.g. on a particular WPS instance, the implementer may still extend the contract of the implementation profile (e.g. by allowing more or larger input data).

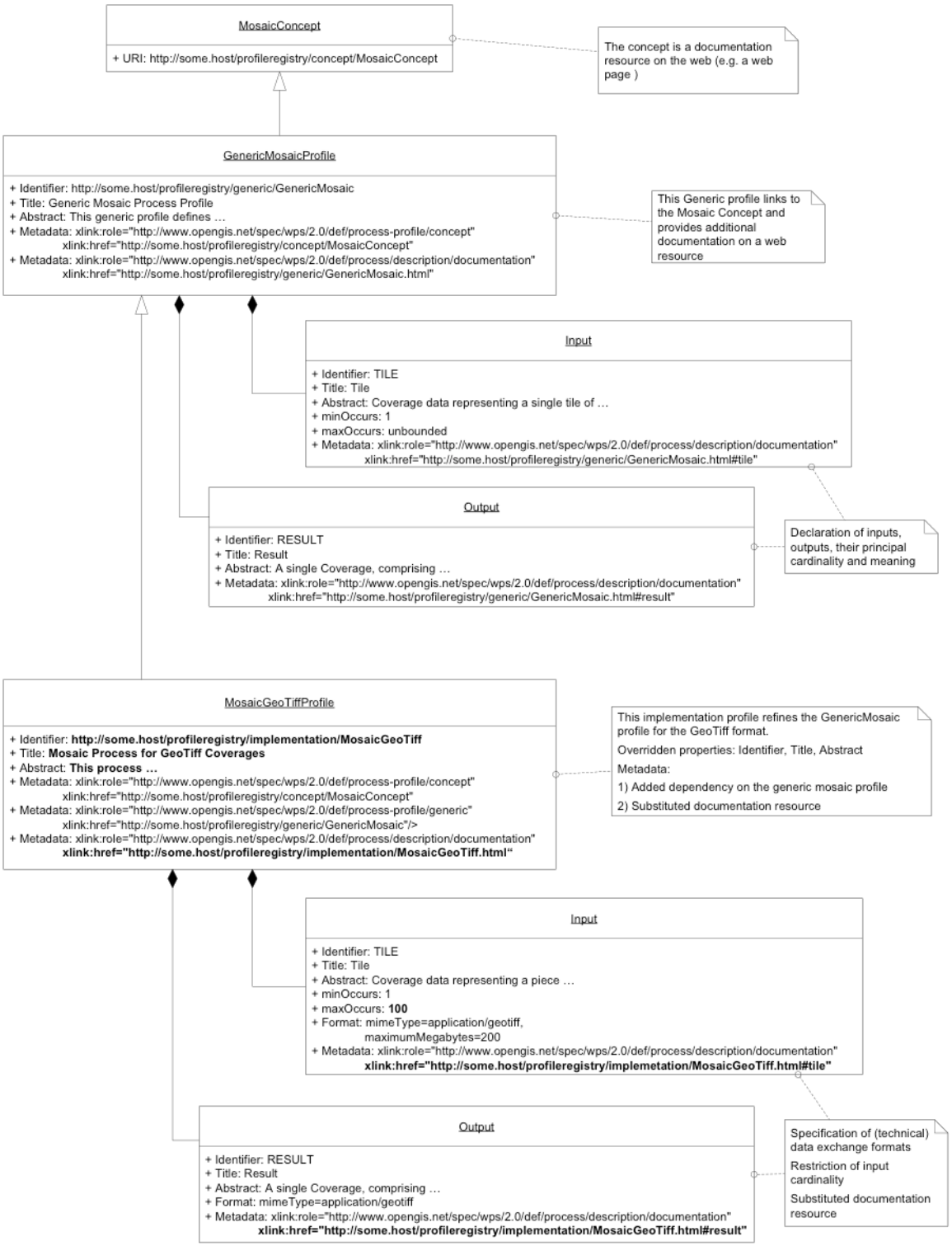

Figure B.1 — Profile inheritance example for a mosaic process

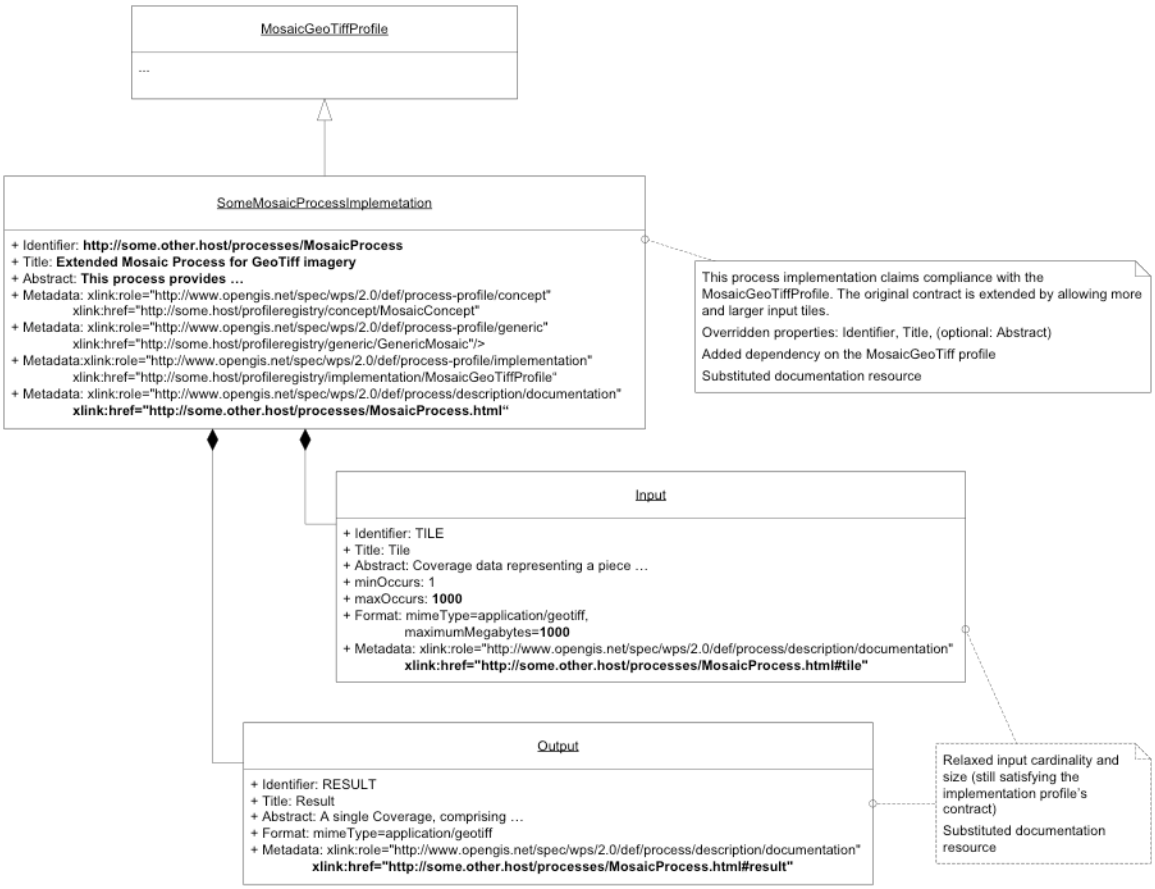

Figure B.2 — Profile inheritance example for a mosaic process, extension by an implementation